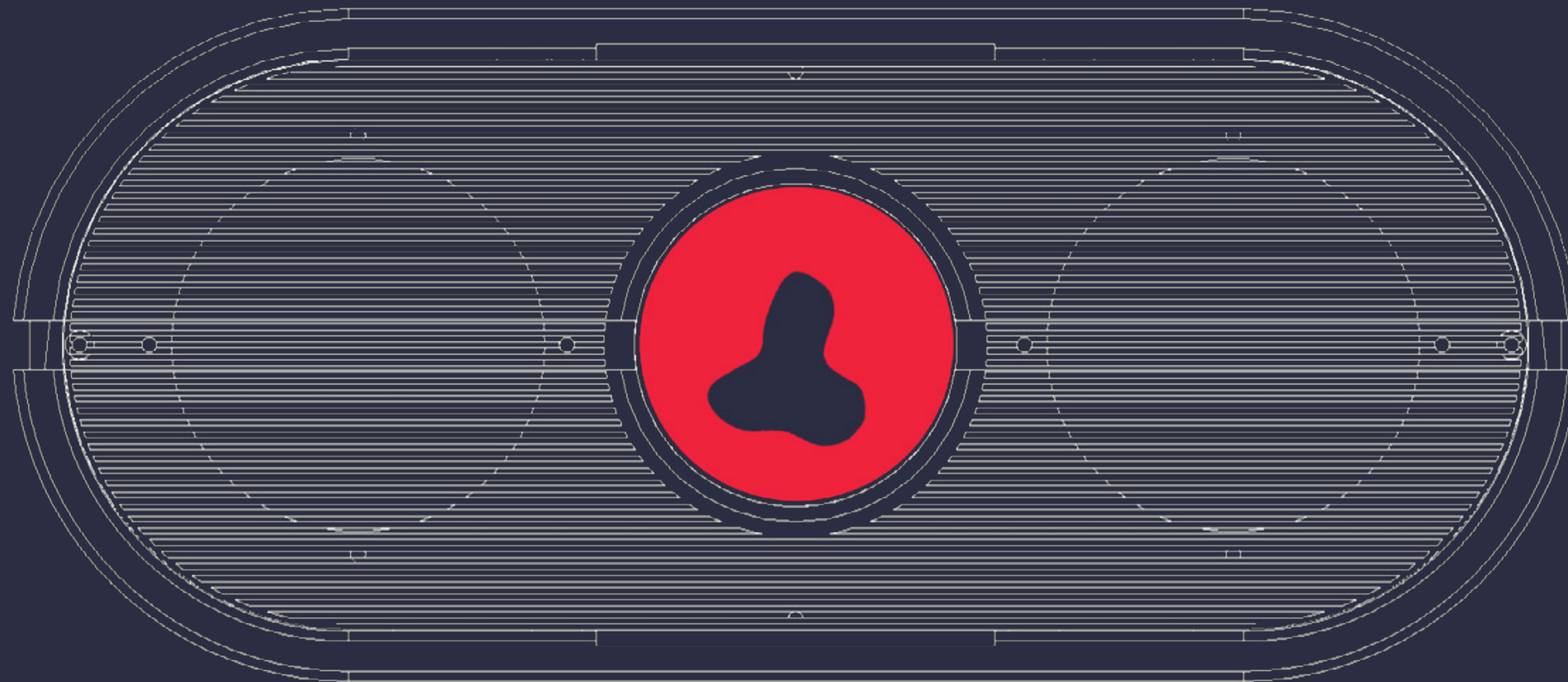


FerroWave

A community funded, open-source, DIY music player with ferrofluid visualizer



[BUILD MANUAL v1.1](#)

FerroWave

FerroWave is a portable, battery-powered device that transforms audio into stunning visual art using ferrofluid - a magnetic liquid that dances and morphs in response to electromagnetic fields.

Connect via Bluetooth or AUX cable, play your favorite music, and watch as the ferrofluid creates hypnotic patterns synchronized perfectly with the beats, bass, and rhythm.

Key Features

Dual Audio Input

- Bluetooth A2DP - Wireless streaming from any phone or computer
- AUX/Line-In - 3.5mm audio input with auto-detection
- Seamless switching between sources

8 Ferrofluid Behavior Modes

- SMOOTH - Gentle, flowing waves
- SPIKE - Sharp, aggressive spikes
- BOUNCE - Rhythmic pumping motion
- CHAOS - Unpredictable, maximum drama
- PULSE - Beat-synchronized pulses
- WAVE - Slow building surges
- TREMOLO - Rapid flutter effect
- BREATH - Meditative breathing

10 LED Visual Effects

- Rainbow Spin
- Spectrum Analyzer
- VU-Meter
- Pulse & Strobe
- Fire & Ocean Waves
- And other experimental ones

8 EQ Presets

- Flat, Bass Boost, Treble
- Vocal, Rock, Electronic
- Jazz, Classical

Six Physical Buttons

- Cycle magnet modes (up/down)
- Change LED effects (up/down)
- Switch EQ presets (up/down)

Portable Design

- 4x18650 battery powered
- cca 2 hours runtime
- Rechargeable via Micro-USB
- Truly cordless operationv

Open Source & Community

FerroWave is 100% open source and was made possible by the amazing MakerWorld community through crowdfunding. Thank you to all the MakerWorld backers who supported the campaign, BambuLab and Makerworld platform for enabling this crowdfunding maker innovation, open-source Arduino and ESP32 communities, library authors: pschatzmann (AudioTools, ESP32-A2DP), Adafruit (NeoPixel).

Get Involved

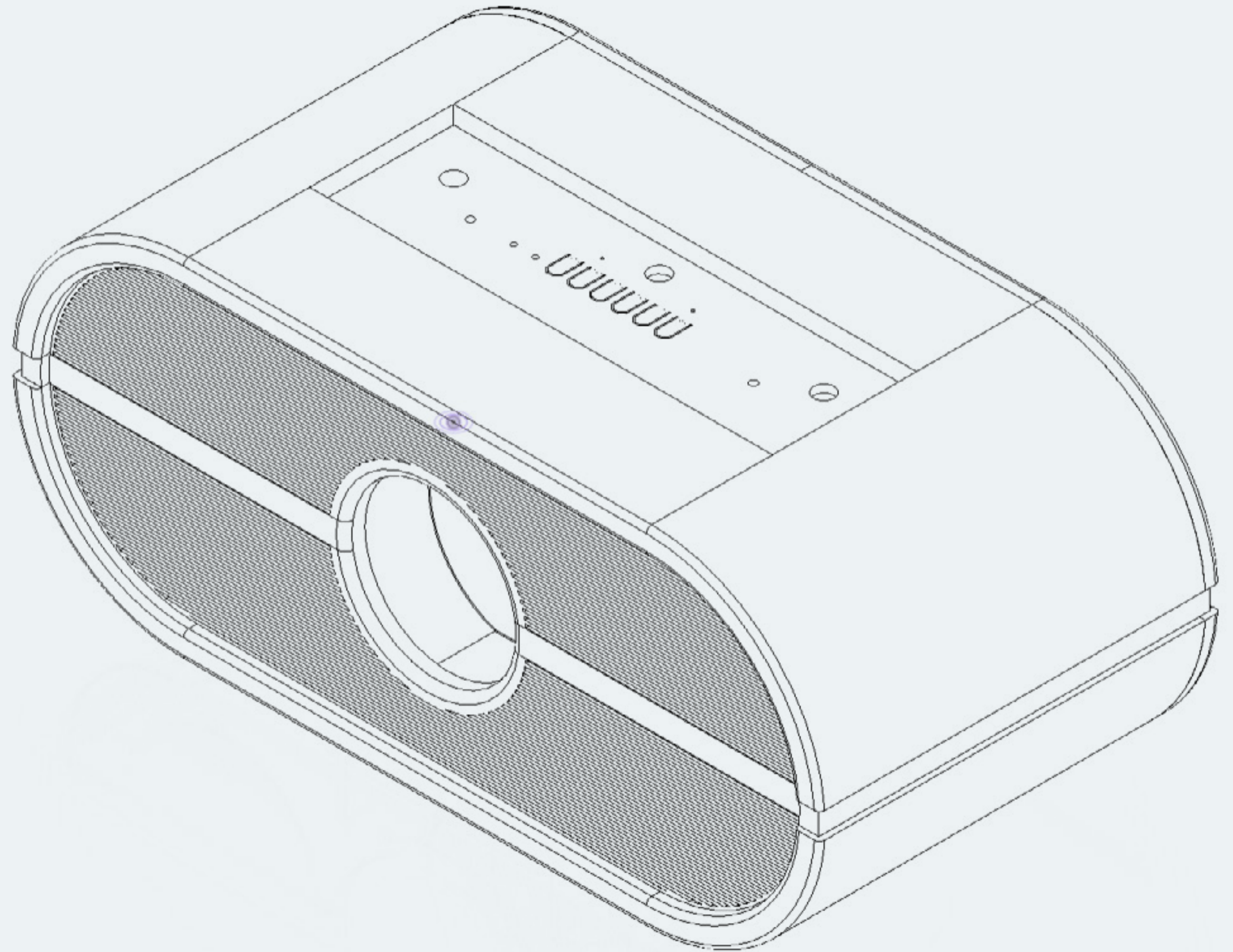
- GitHub: github.com/matoslav/FerroWave
- MakerWorld: makerworld.com/en/crowdfunding/70-ferrowave

Created by: Makarov87

Crowdfunded by: MakerWorld Community

License: MIT Open Source

Special thanks to everyone who contributed ideas, testing, and feedback! Turn the page to begin building your FerroWave...

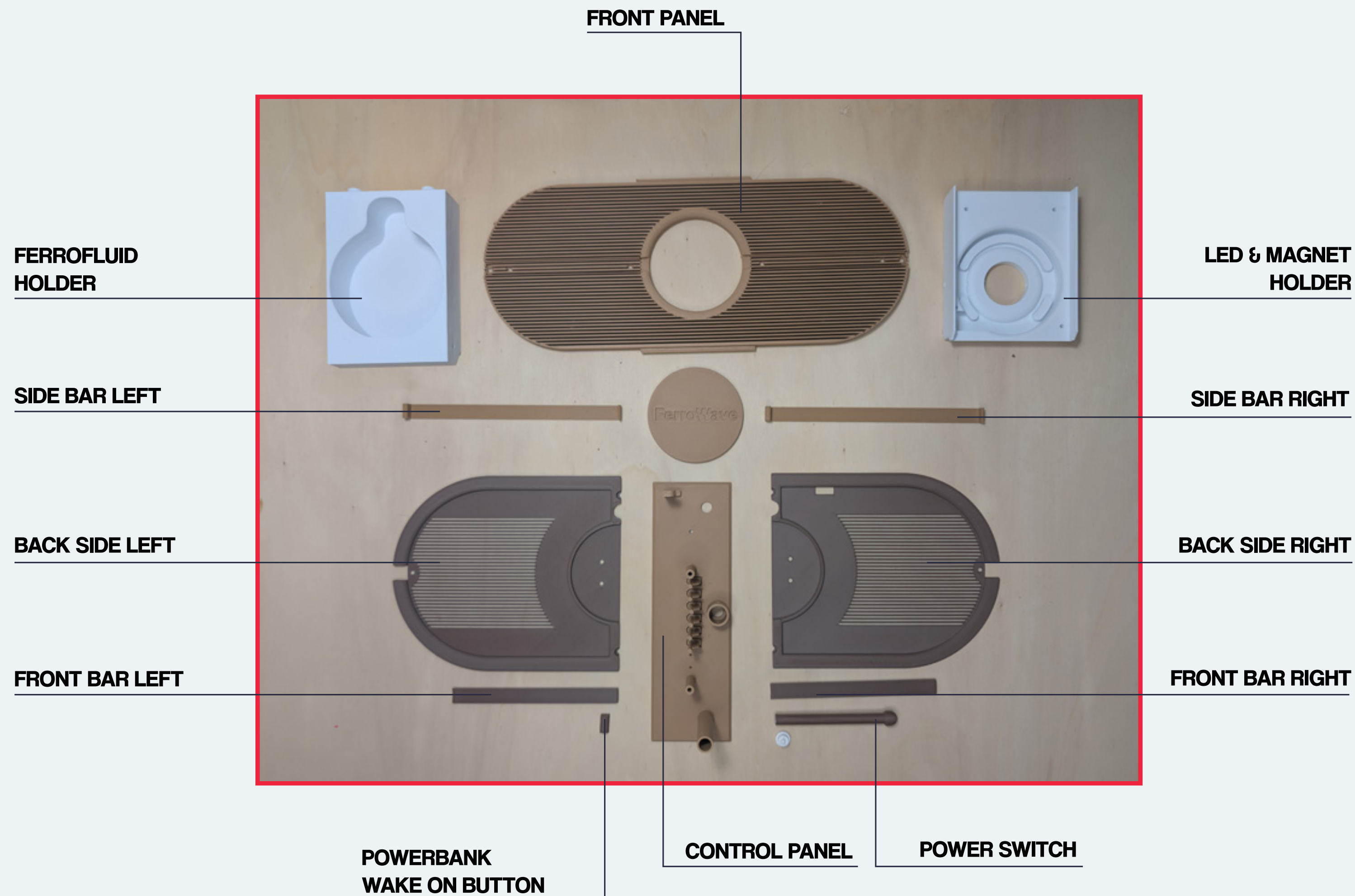


1

LIST OF MATERIALS

1.1

3D PRINTED PARTS



1

LIST OF MATERIALS

1.2

ELECTRONIC COMPONENTS

BATTERY SHIELD 4X18650

[Amazon.de link for the exact ones I used](#)

ESP32-A1S AUDIO KIT BOARD BY AITHINKER

[Amazon.de link for the exact ones I used](#)

SPEAKERS

[Amazon.de link for the exact ones I used](#)

10W AUDIO POWER AMP

[Amazon.de link for the exact ones I used](#)

2X NEODYMIUM MAGNETS

[Amazon.de link for the exact ones I used](#)

24-BIT LED RING (NEOPIXEL)

[Amazon.de link for the exact ones I used](#)

5V ELECTROMAGNET

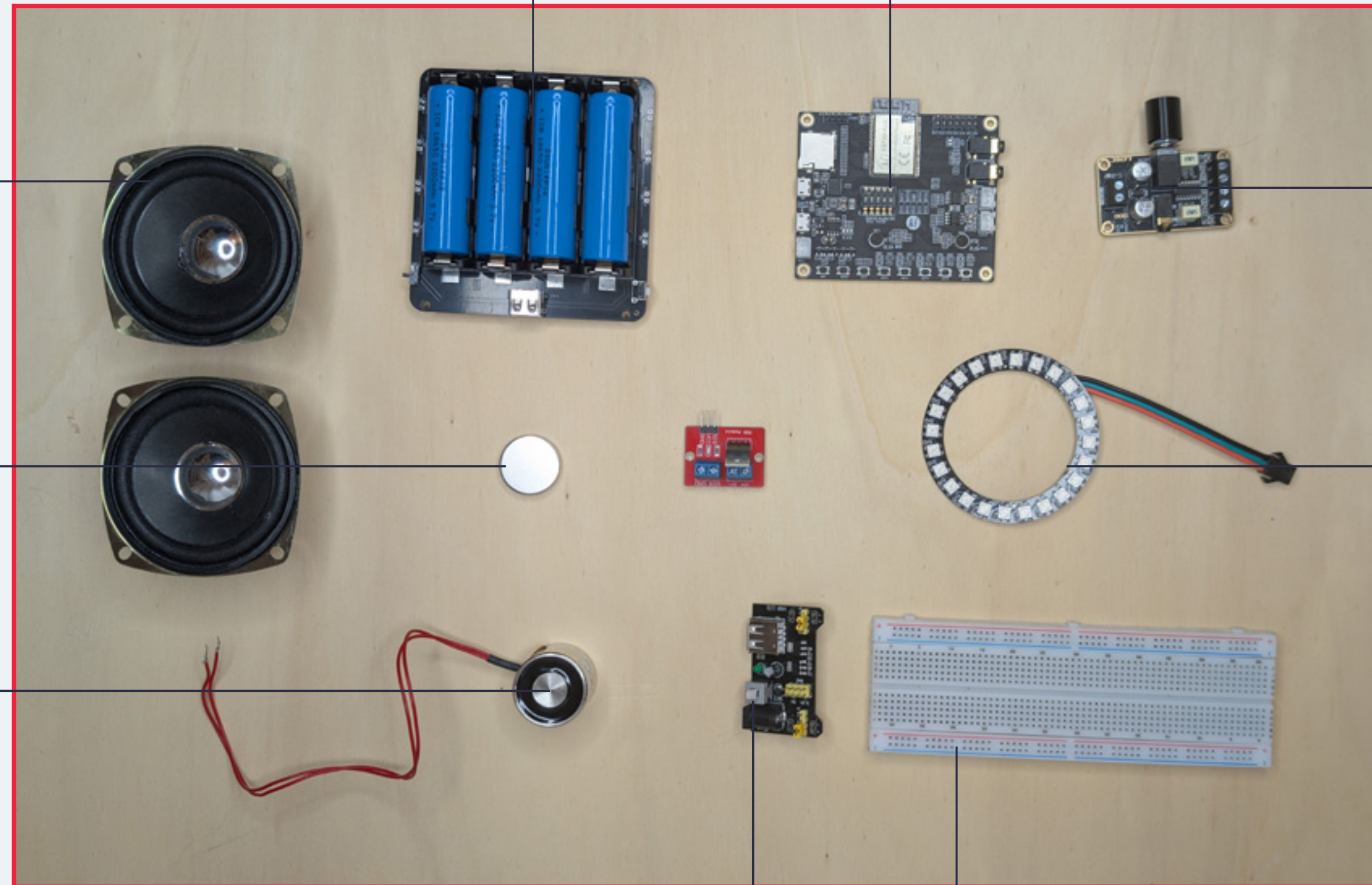
[Amazon.de link for the exact ones I used](#)

BREADBOARD POWER MODULE

[Amazon.de link for the exact ones I used](#)

BREADBOARD

[Amazon.de link for the exact ones I used](#)



1

LIST OF MATERIALS

1.3

OTHER



Ferrofluid in a bottle

[Amazon.de link for the exact ones I used](#)

OR YOU CAN MIX
YOUR OWN:



Ferrofluid separate

[Amazon.de link for the exact ones I used](#)



Silicone Oil

[Amazon.de link for the exact ones I used](#)

More detailed instructions on how to
make your own will follow soon.

1

LIST OF MATERIALS

1.3

OTHER



M3 screws:

- 6mm 24x
- 8mm 4X
- 12mm 2x
- 16mm 2x



Short mini jack cable

Female - male



Male - male



Dupont cables

- Male to male
- Male to female
- Female to female



USB cables

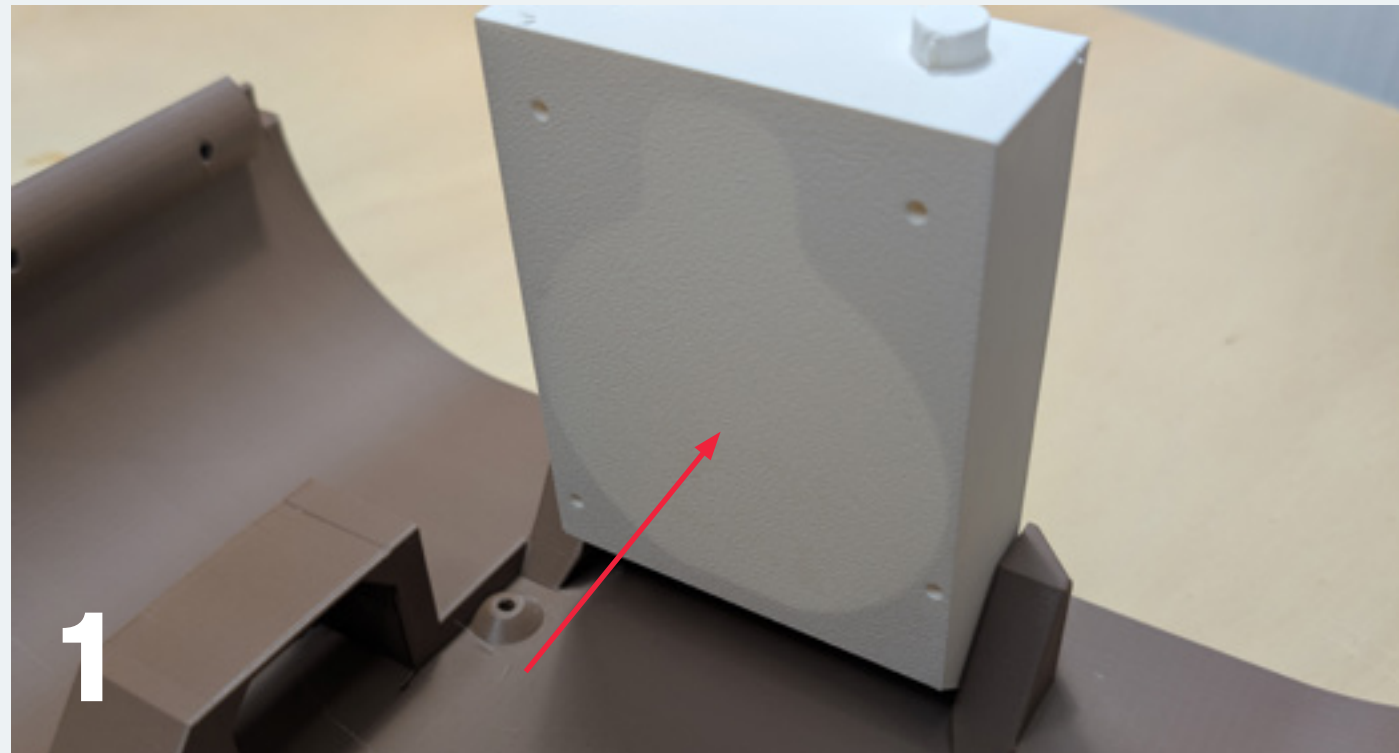
1x USB-A to USB-A (or you can splice and join two existing ones you have at home)
2x micro-USB cables (these will have to be cannibalized)

2

ASSEMBLY

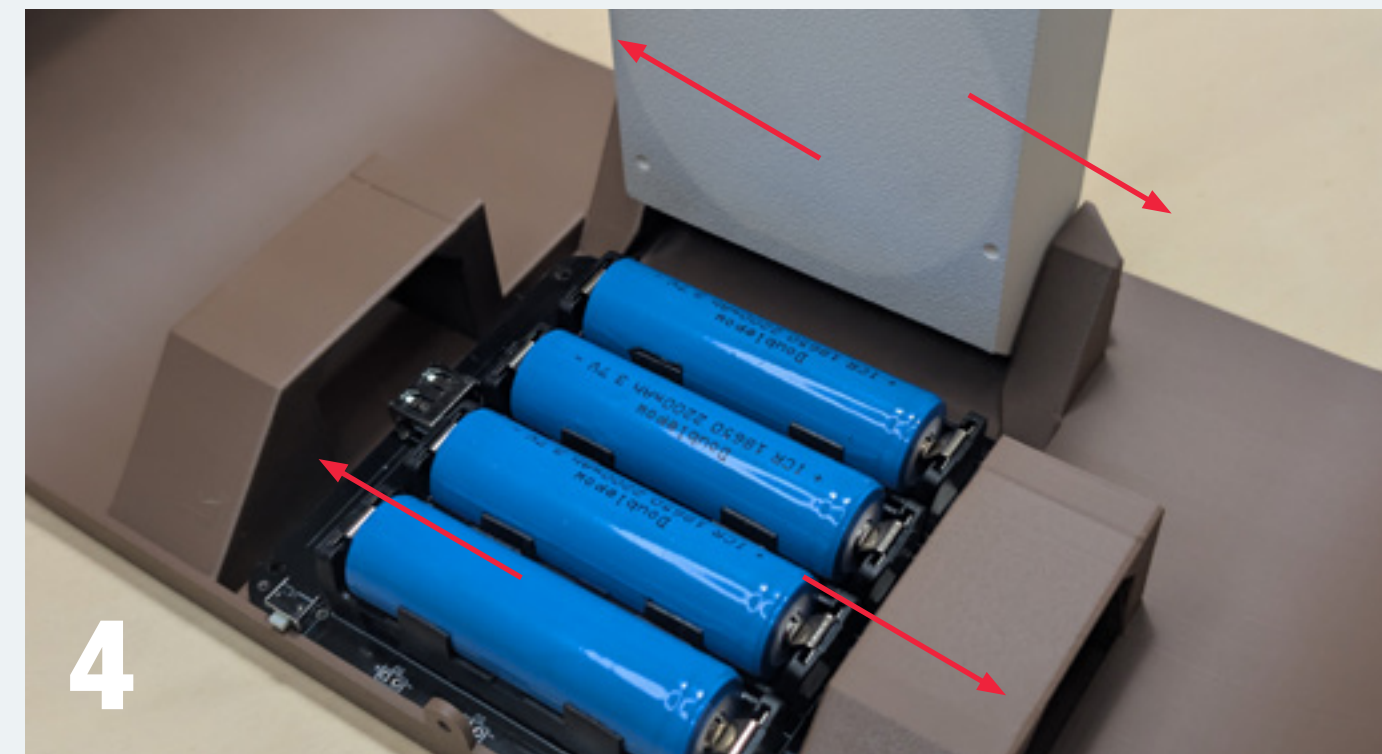
2.1

FERROFLUID HOLDER and BATTERY PACK



Take the Ferrofluid holder and slide it into the slot on the Bottom Panel. Secure them with two M3x12mm screws.

Take the Battery Pack, place it on the Bottom Panel and secure it with four M3x3mm screws.

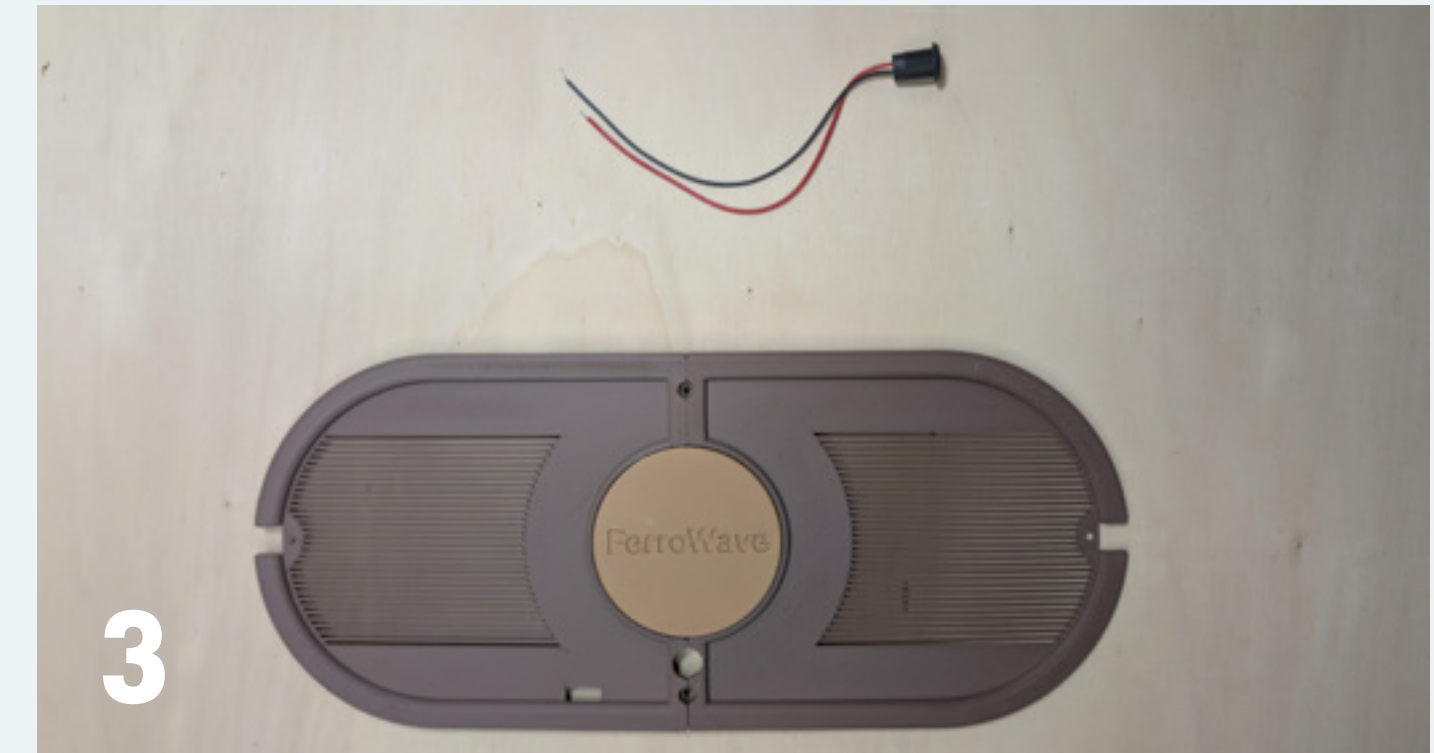
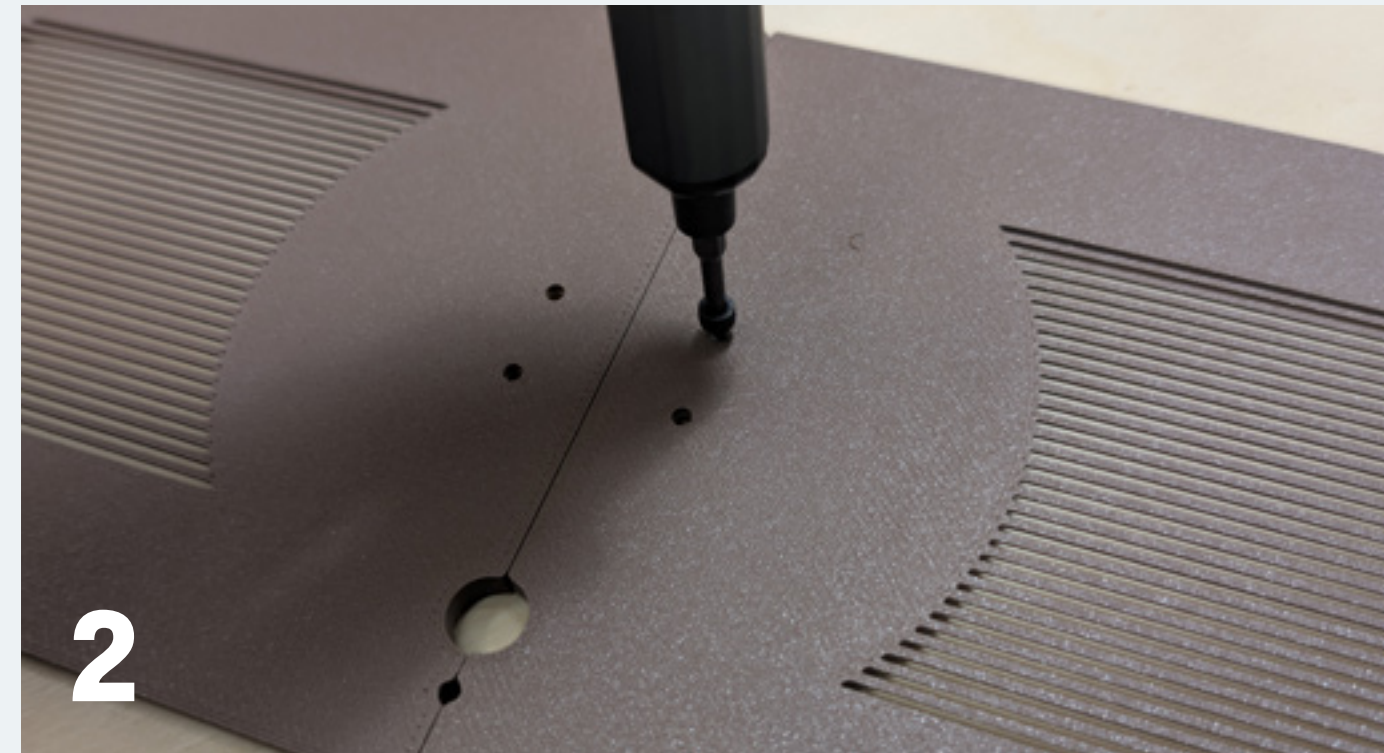


2

ASSEMBLY

2.2

BACK PLATE



To connect the two halves of the Back Plate, place the Back Circle and screw it in place with four M3x3mm screws.

Take the USB-C pigtail cable and secure it in the hole of the Back Panel.

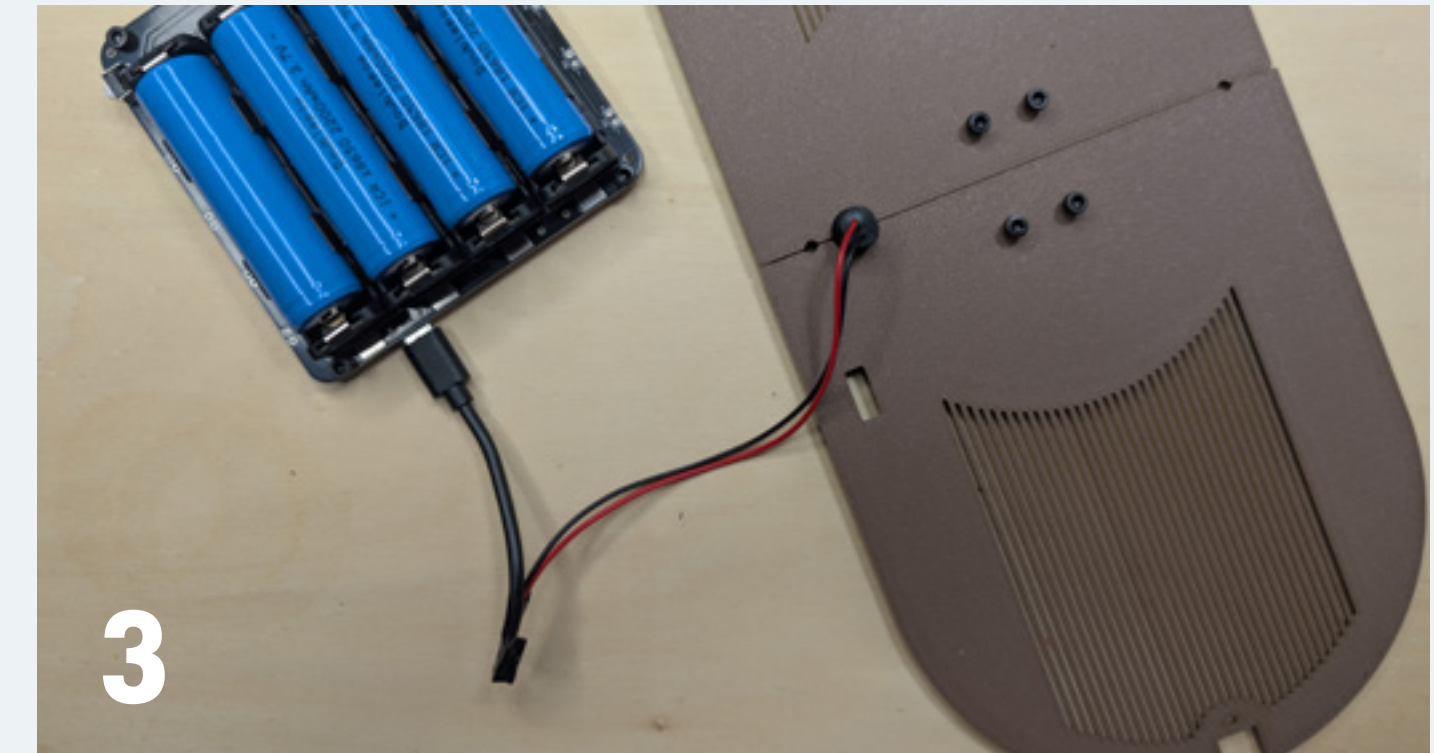
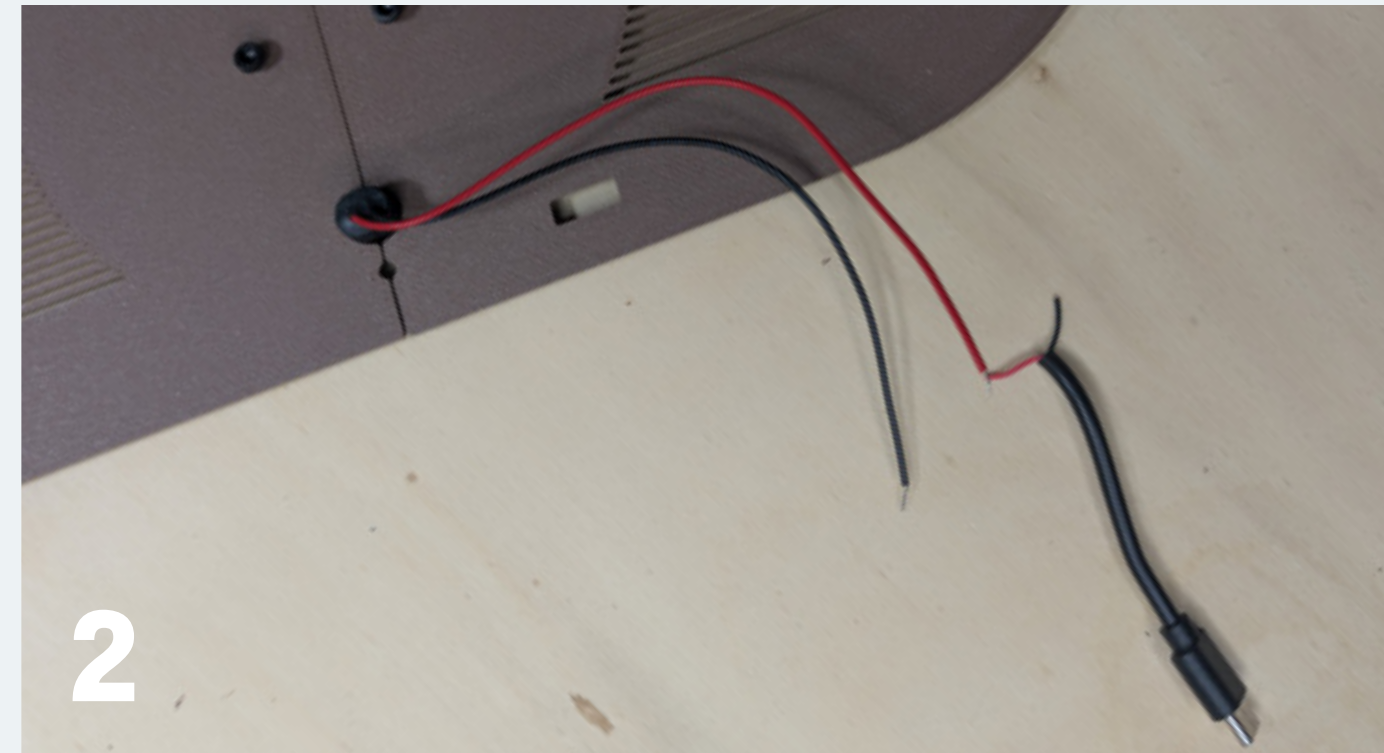
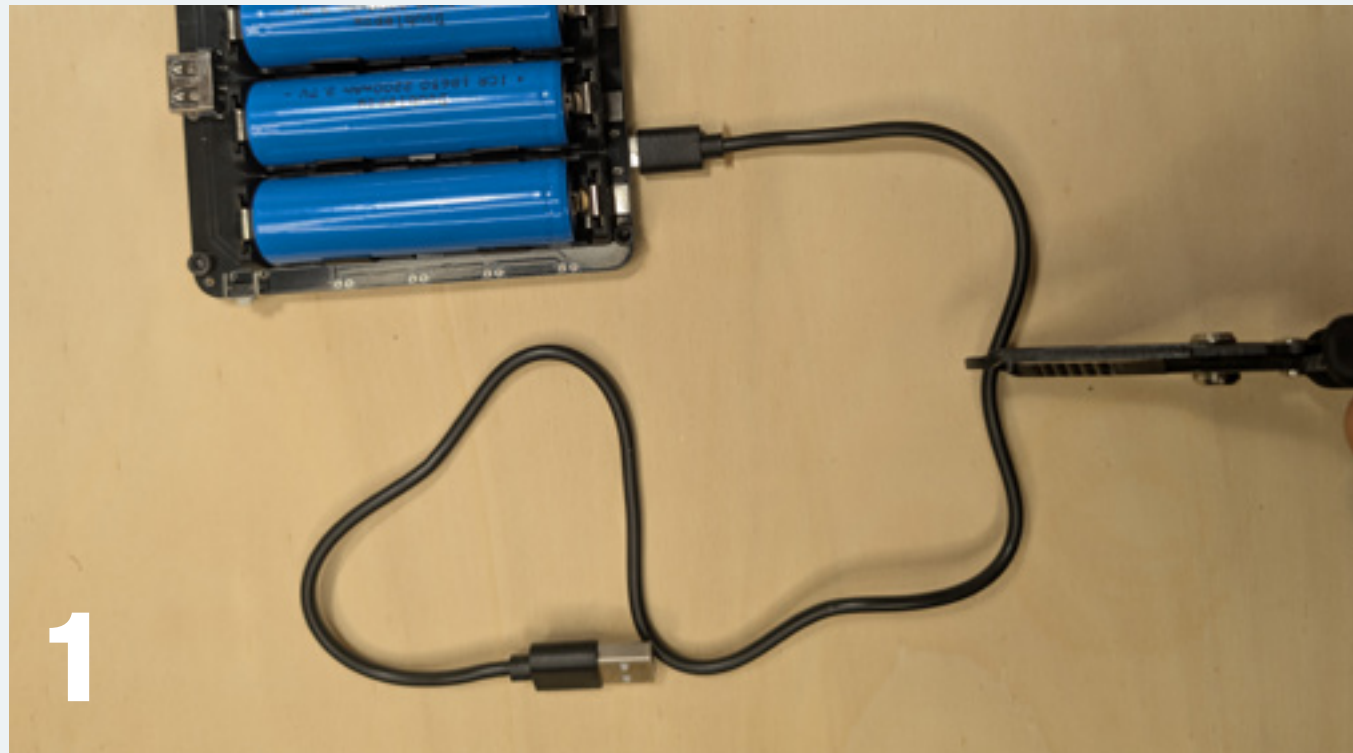


2

ASSEMBLY

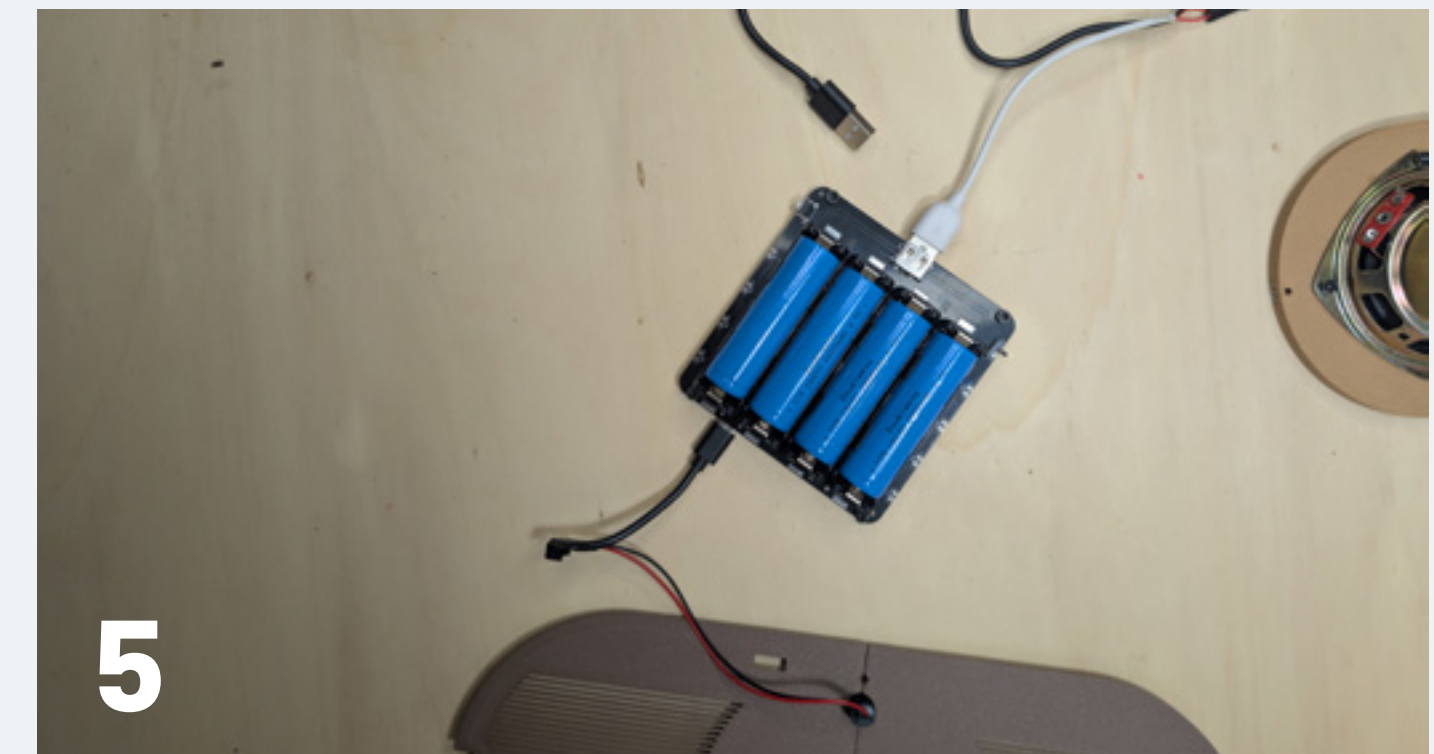
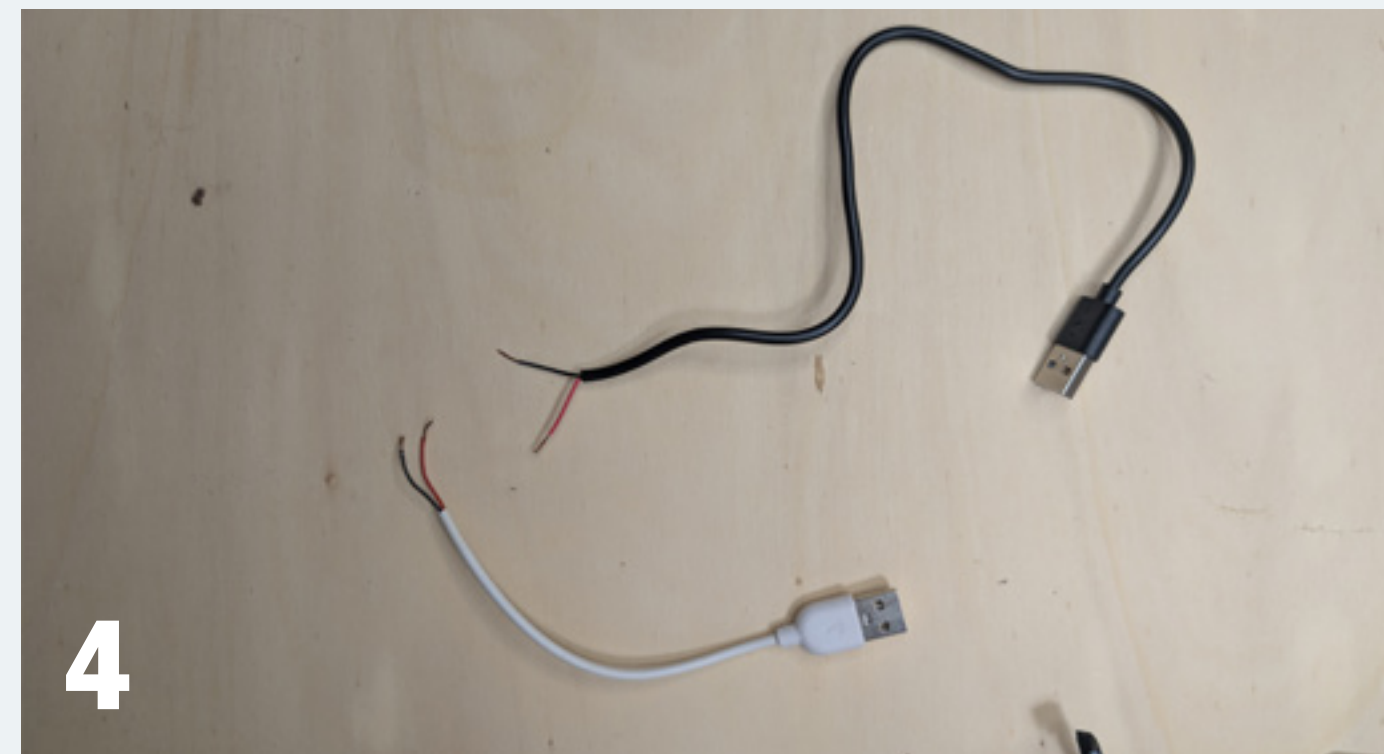
2.2

POWER CABLES



First we need to connect a micro-usb cable to our Back Panel female in. This is for charging the Battery Pack. I cut an existing cable and joined the two together.

We also need to create an USB-A to USB-A cable - you can buy one or make one on your own, like shown here. This will power the breadboard.

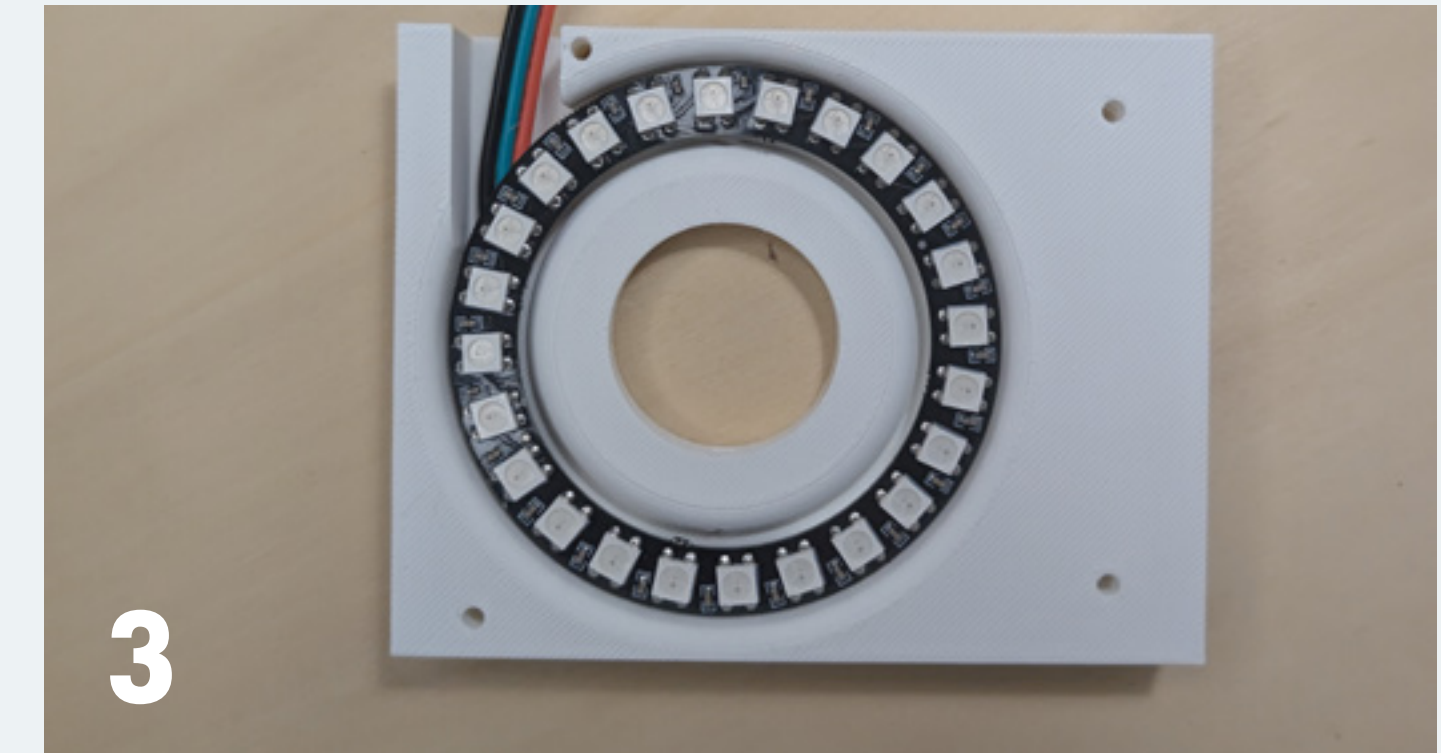
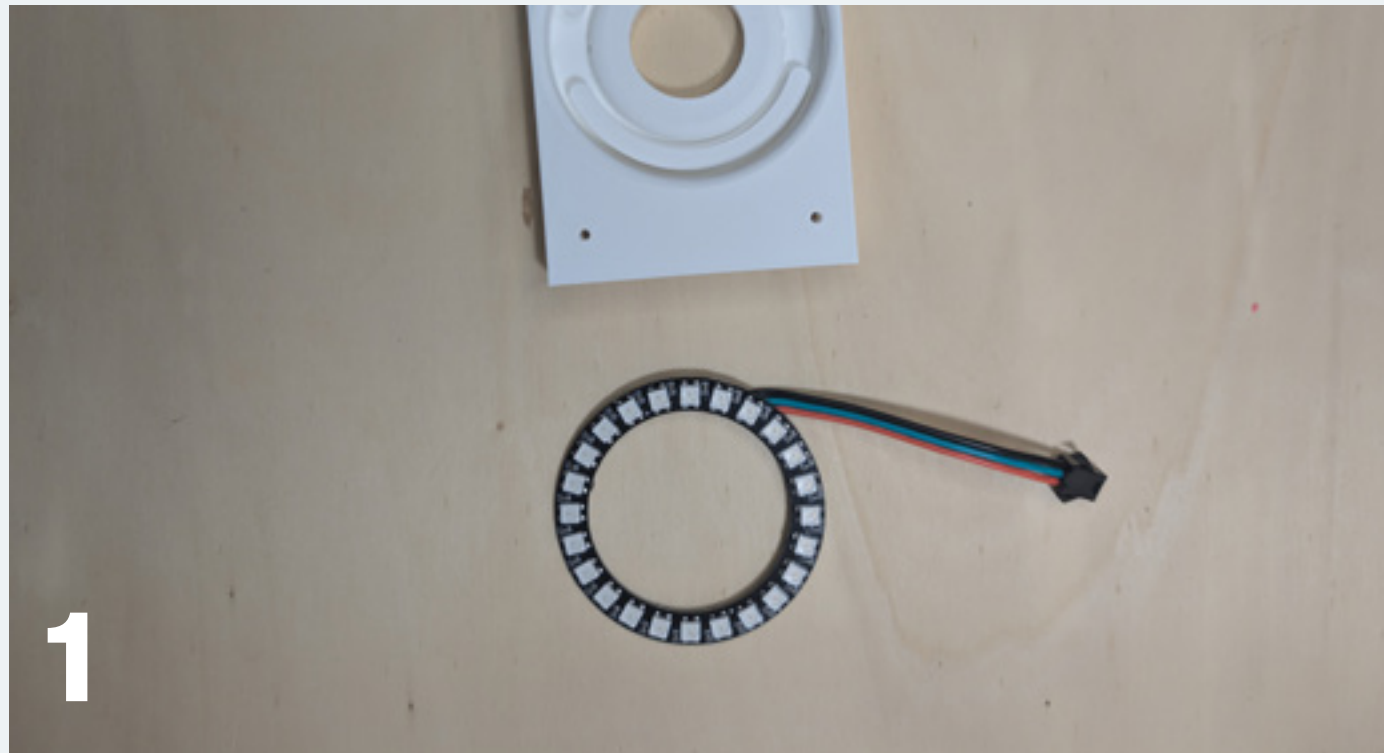


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ASSEMBLY

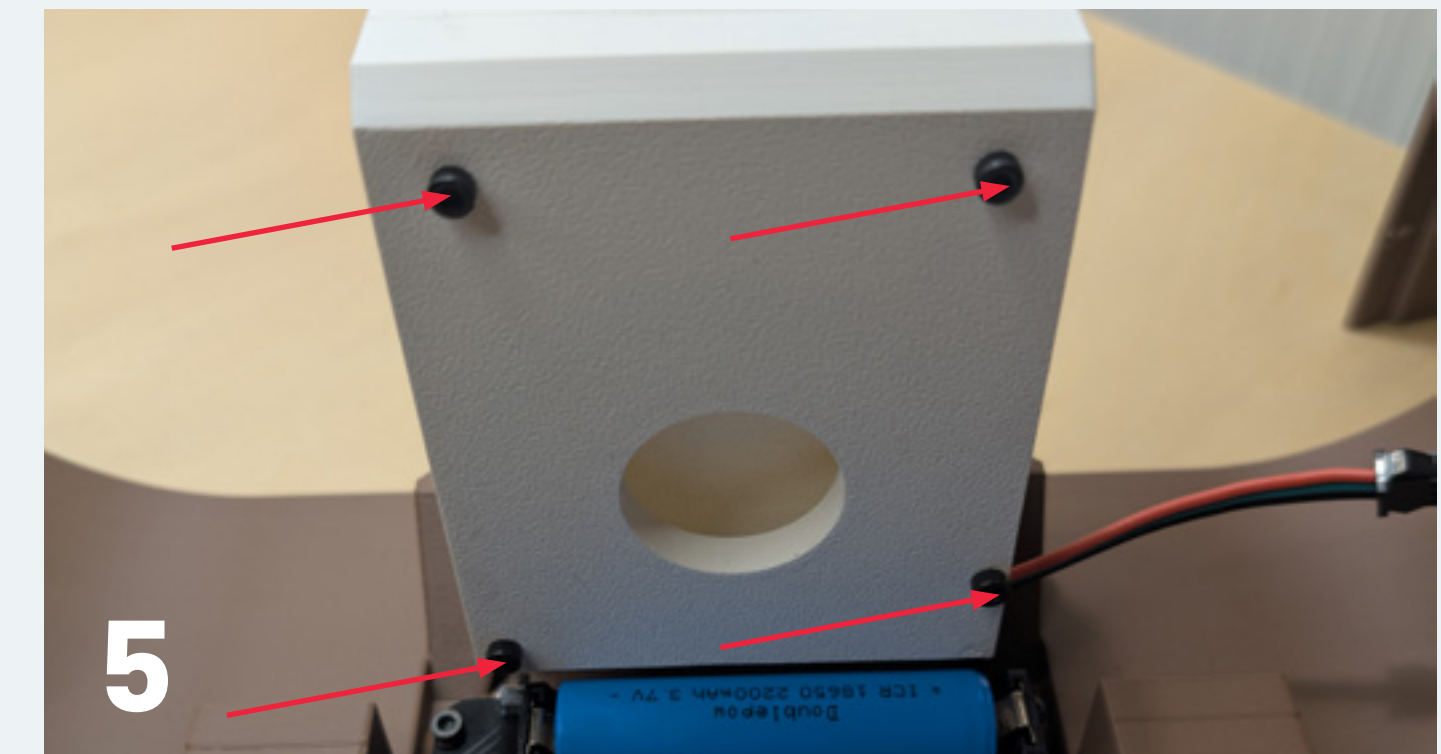
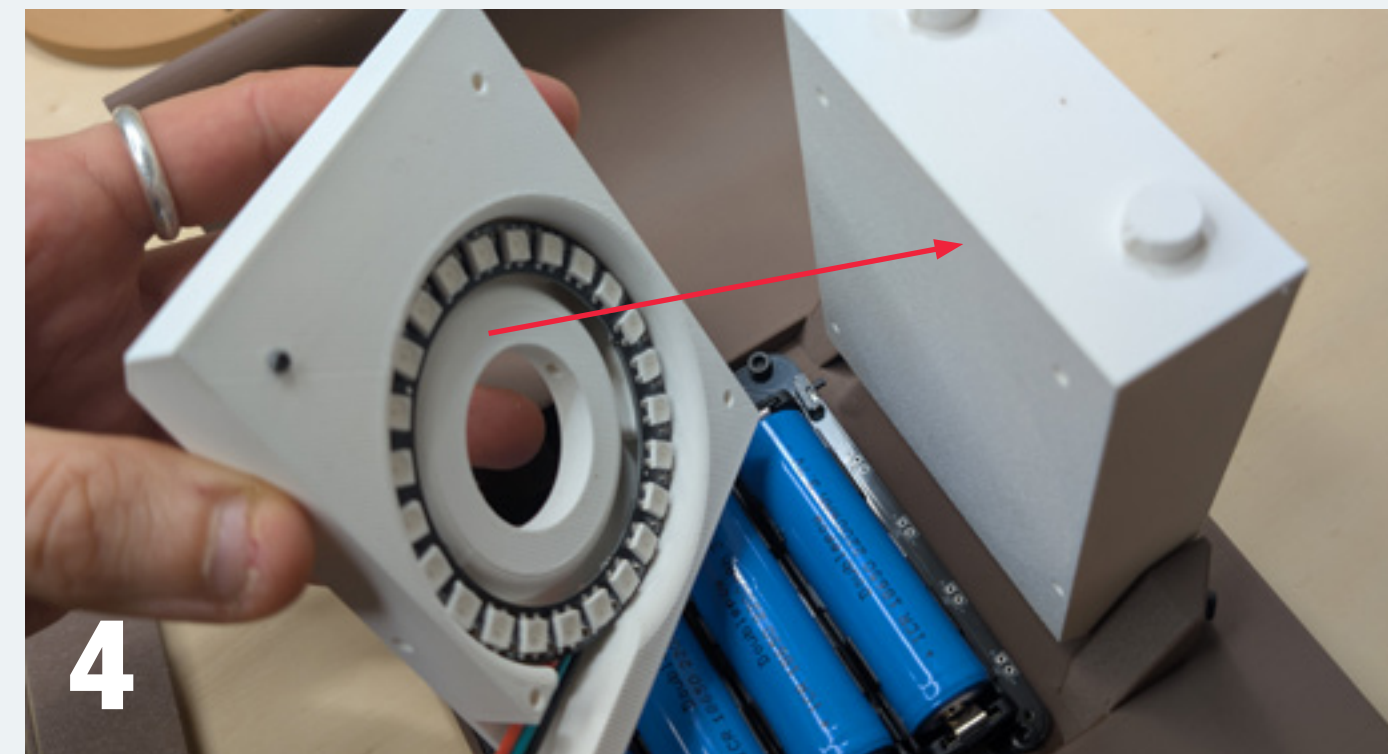
2.2

POWER CABLES



Take the LED Ring - usually it comes with IN and OUT cables - just snip the OUT ones as we dont need them. Glue them onto the base.

Place it on the back of the Ferrofluid holder and screw it in using M3 x 16mm screws.

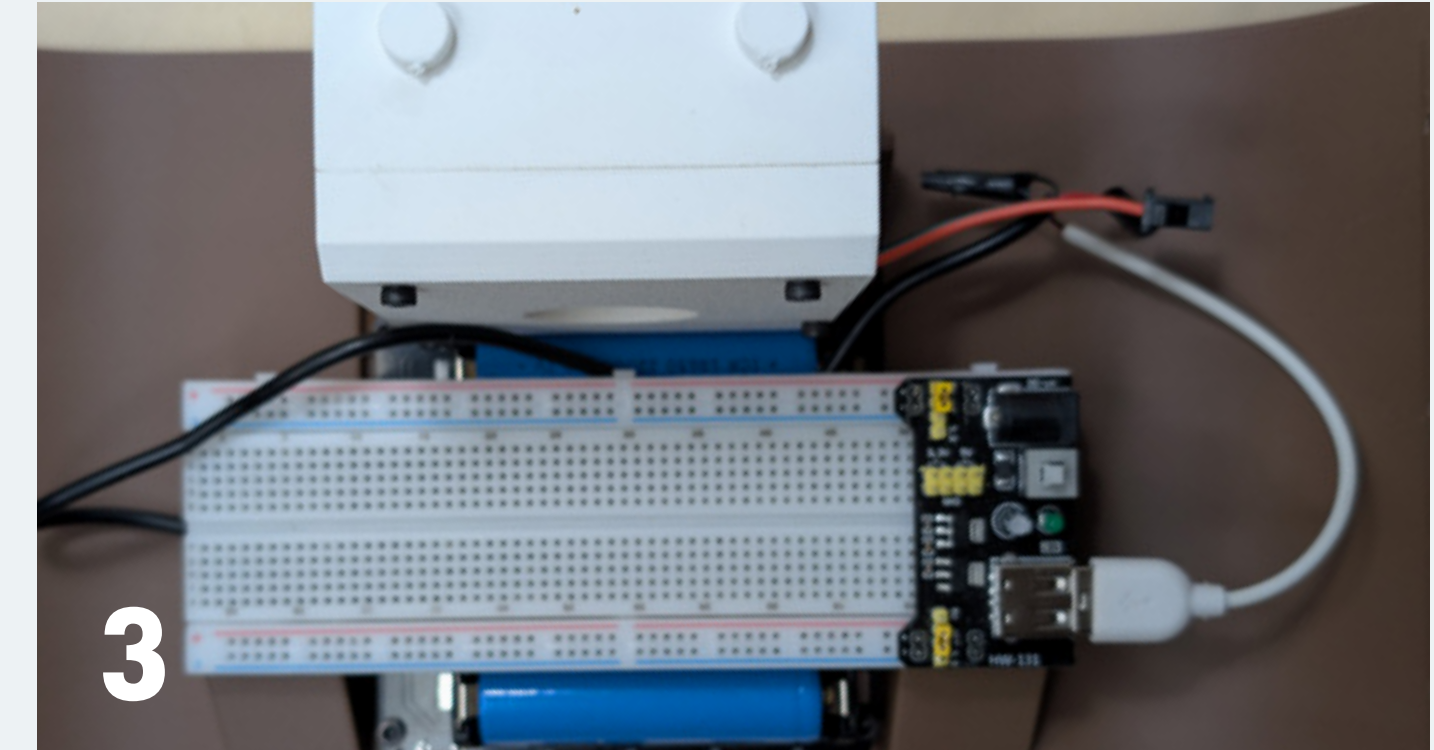
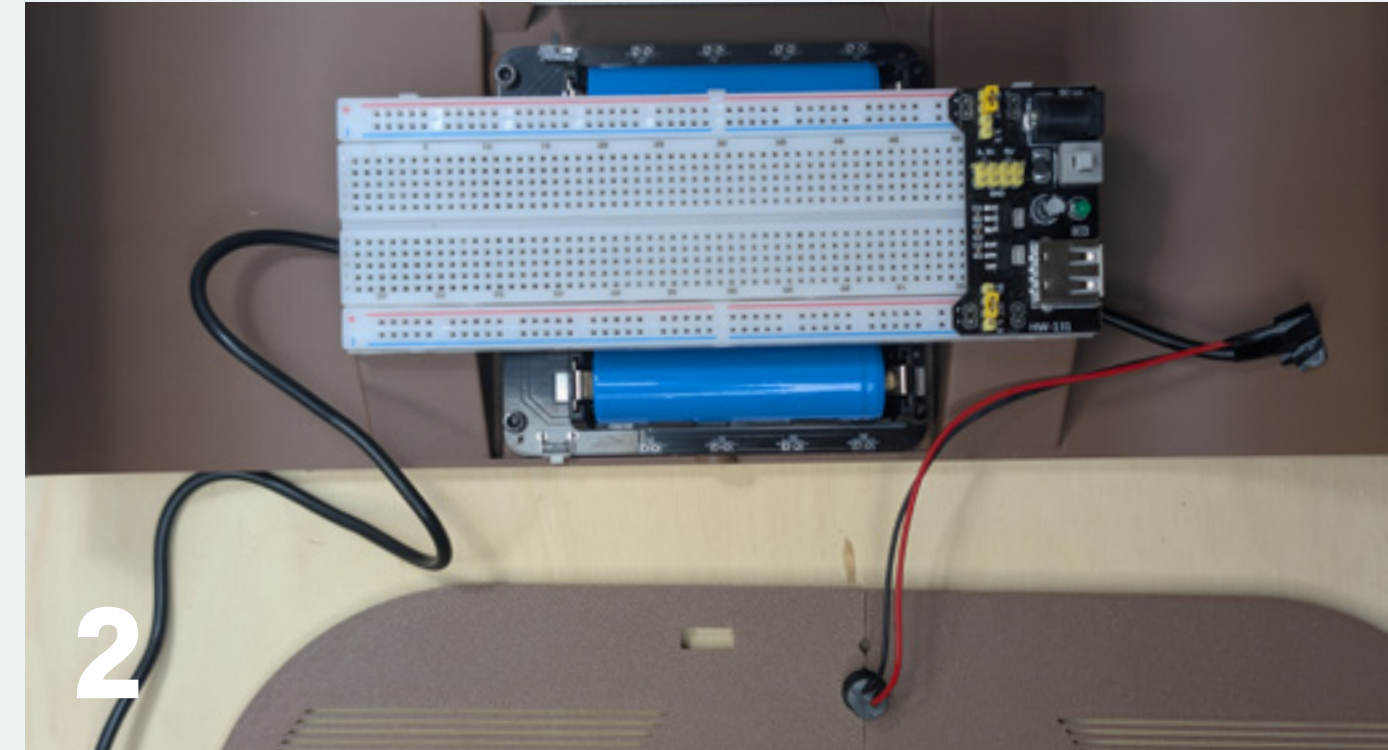
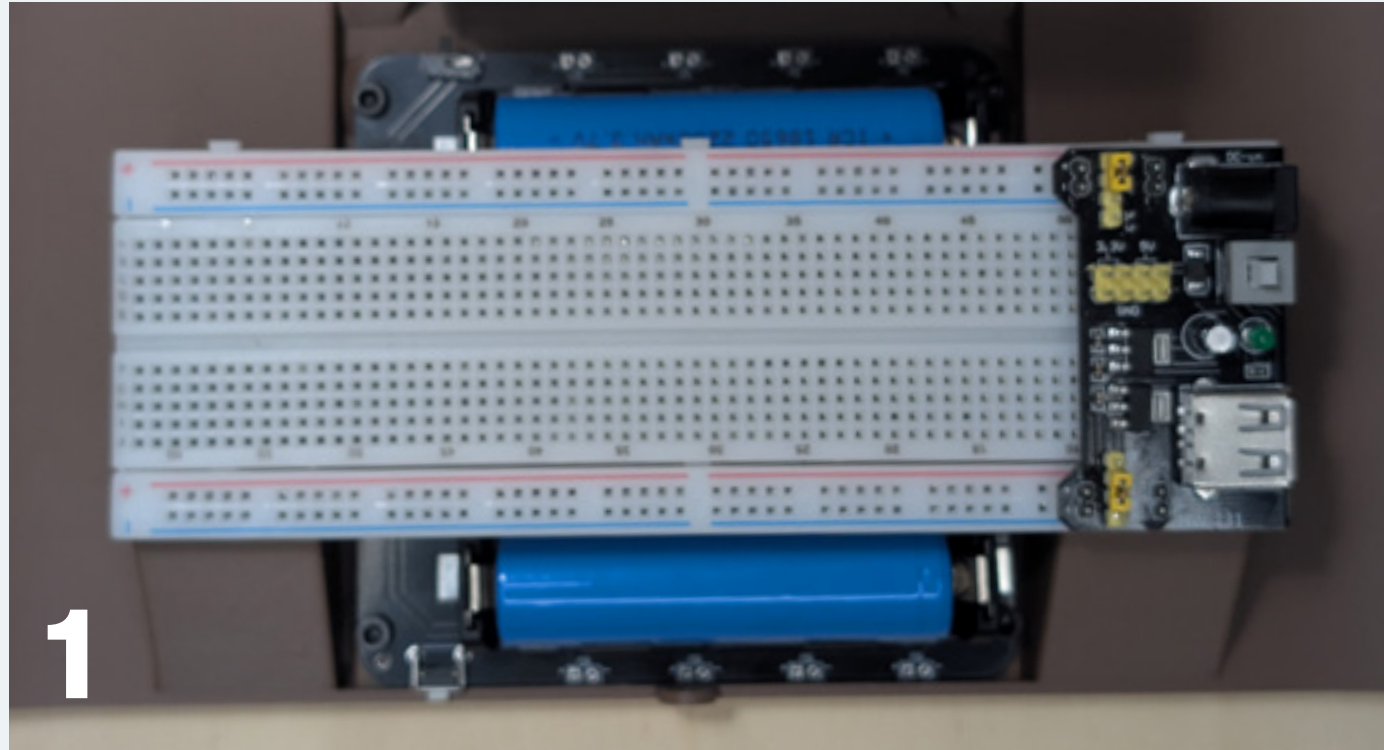


2

ASSEMBLY

2.2

BREADBOARD

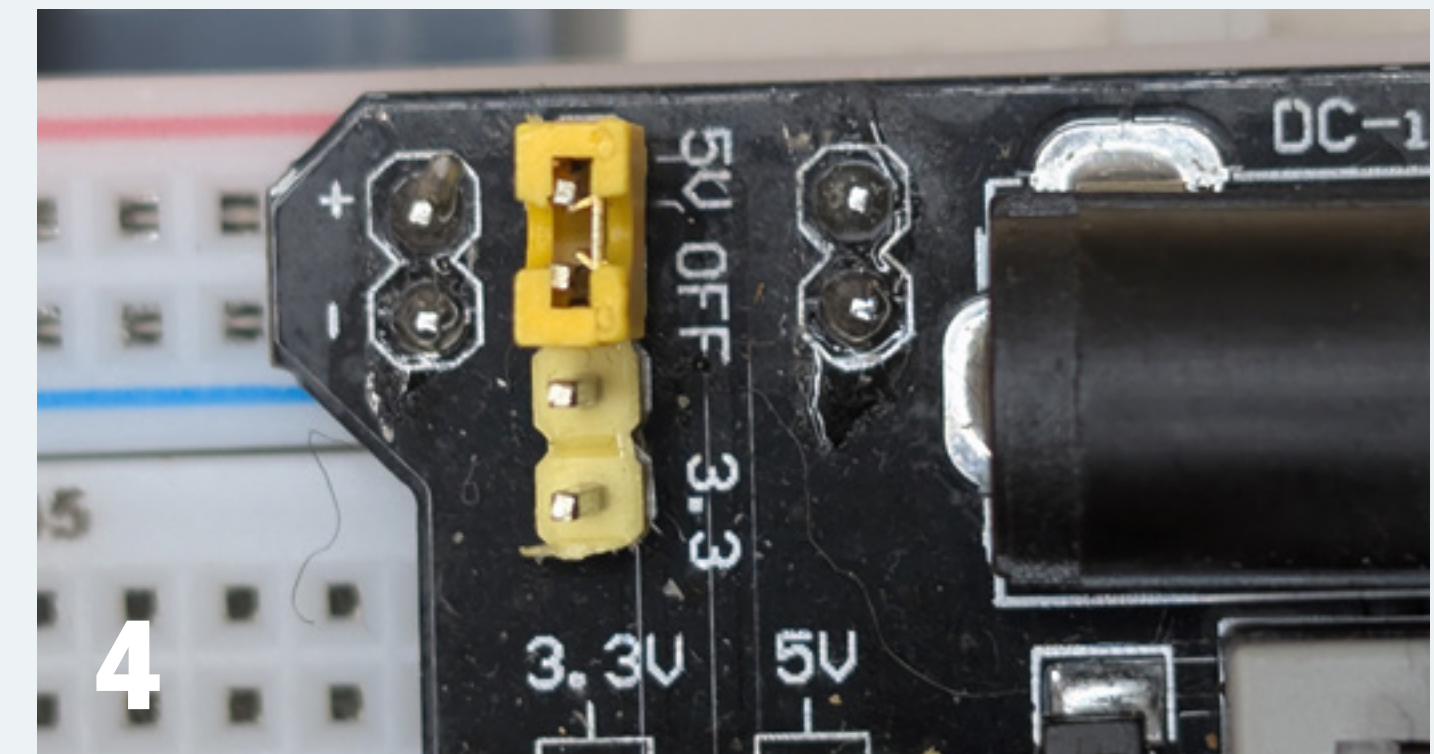


Tape the Breadboard onto the assembly. Also put the Breadboard Power Module on the right hand side.

So on one side of the Battery Pack, micro-USB connects to Back Panel for charging and the USB-A output goes to the Breadboard Power Module USB-A input.

You can disconnect them for now if it makes it easier for assembly and reconnect them later on.

IMPORTANT: Check that the top jumpers are set to 5V!

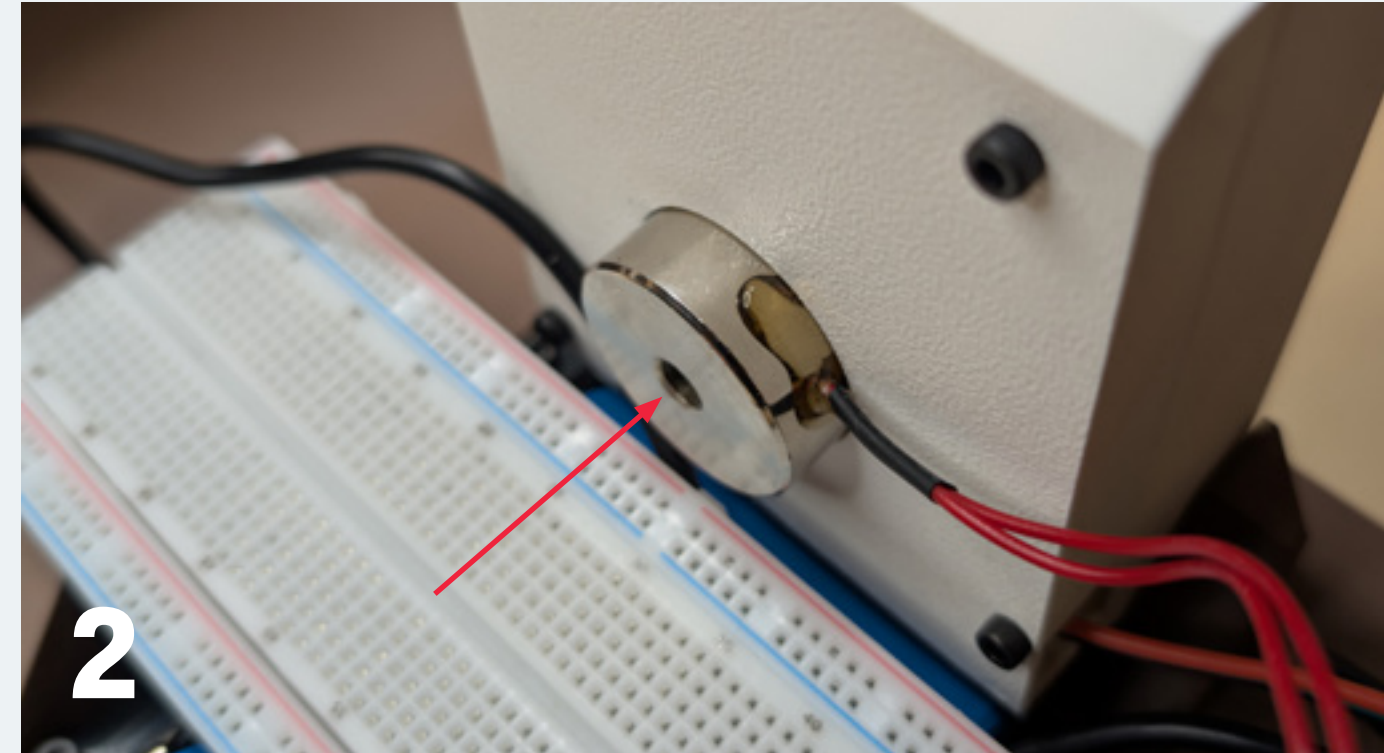
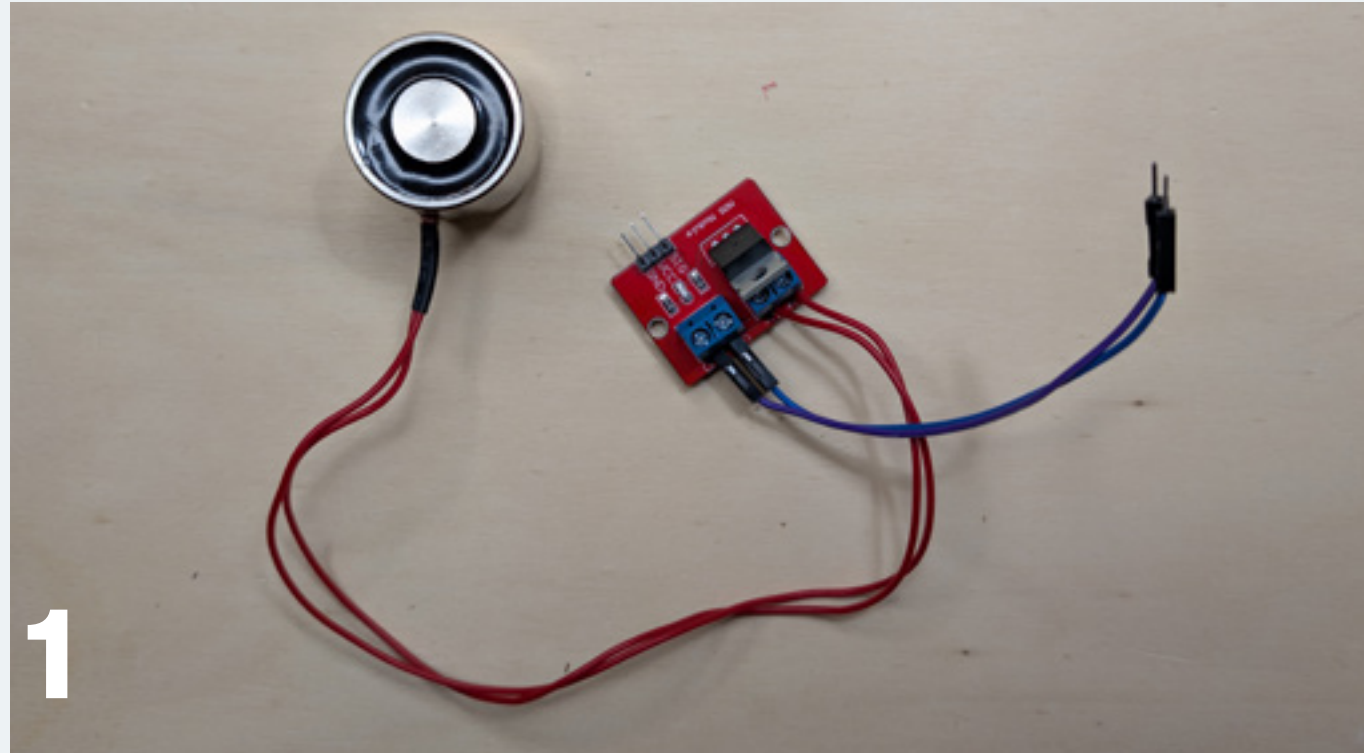


2

ASSEMBLY

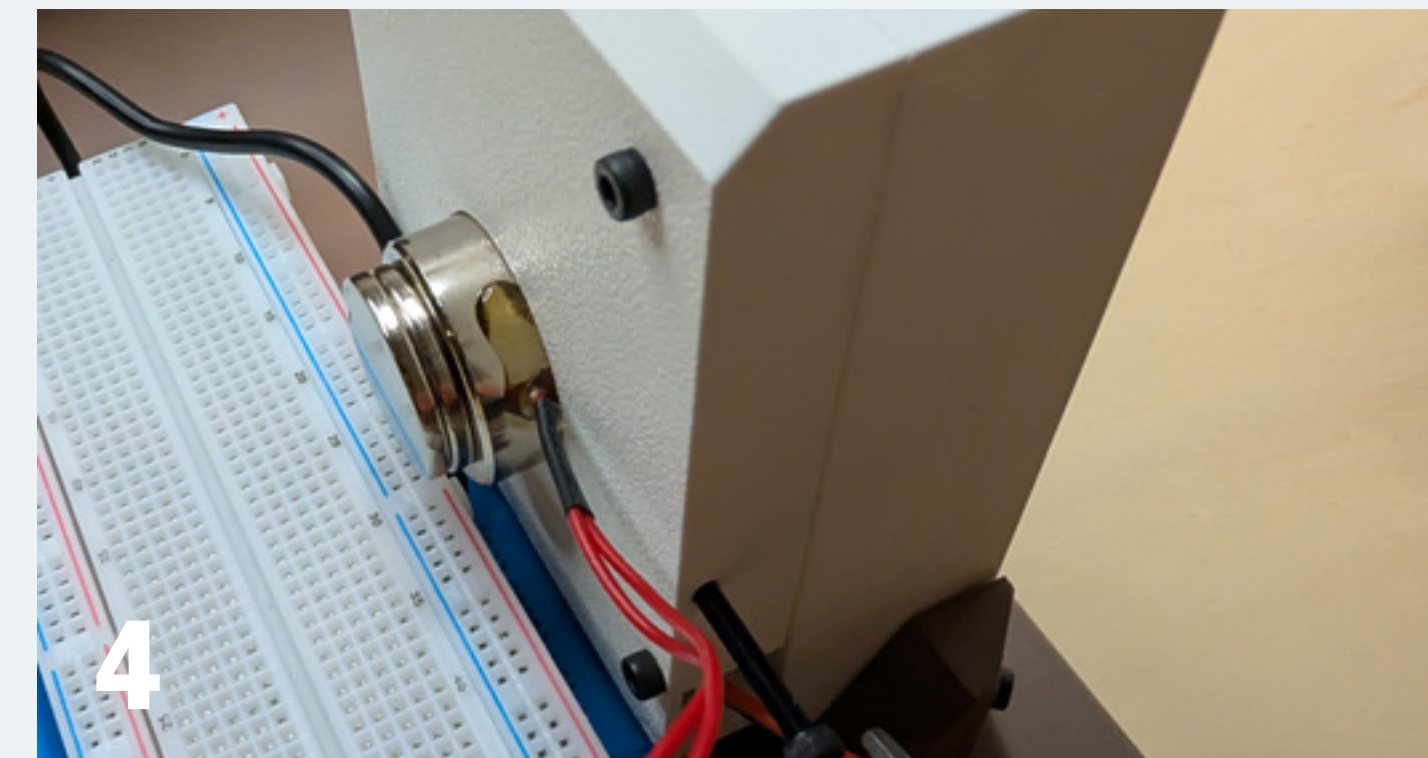
2.2

MOSFET TRIGGER and ELECTROMAGNET



Take the Electromagnet and connect it's two wires with the V+ and V- terminals on the MOSFET modules. Use a screwdriver to tighten them. Take two short male-male Dupont cables, secure them on one end to the MOSFET Vin and GND screw terminals.

Slip the magnet into it's slot. On the sides there are two holes for screws if you wish to more securely fix it in place. On the back of the Electromagnet, place two (or more, experiment) Neodymium magnets. These help with the lift.

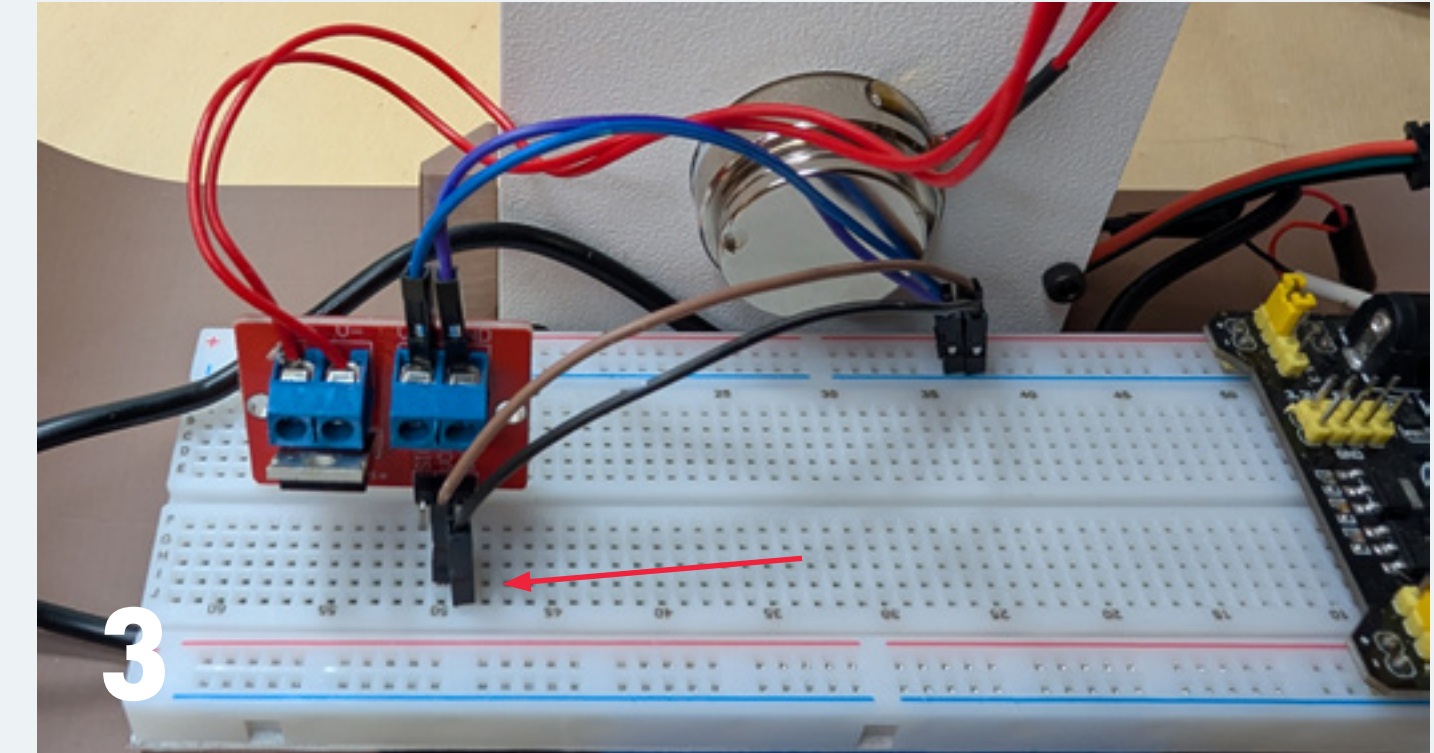
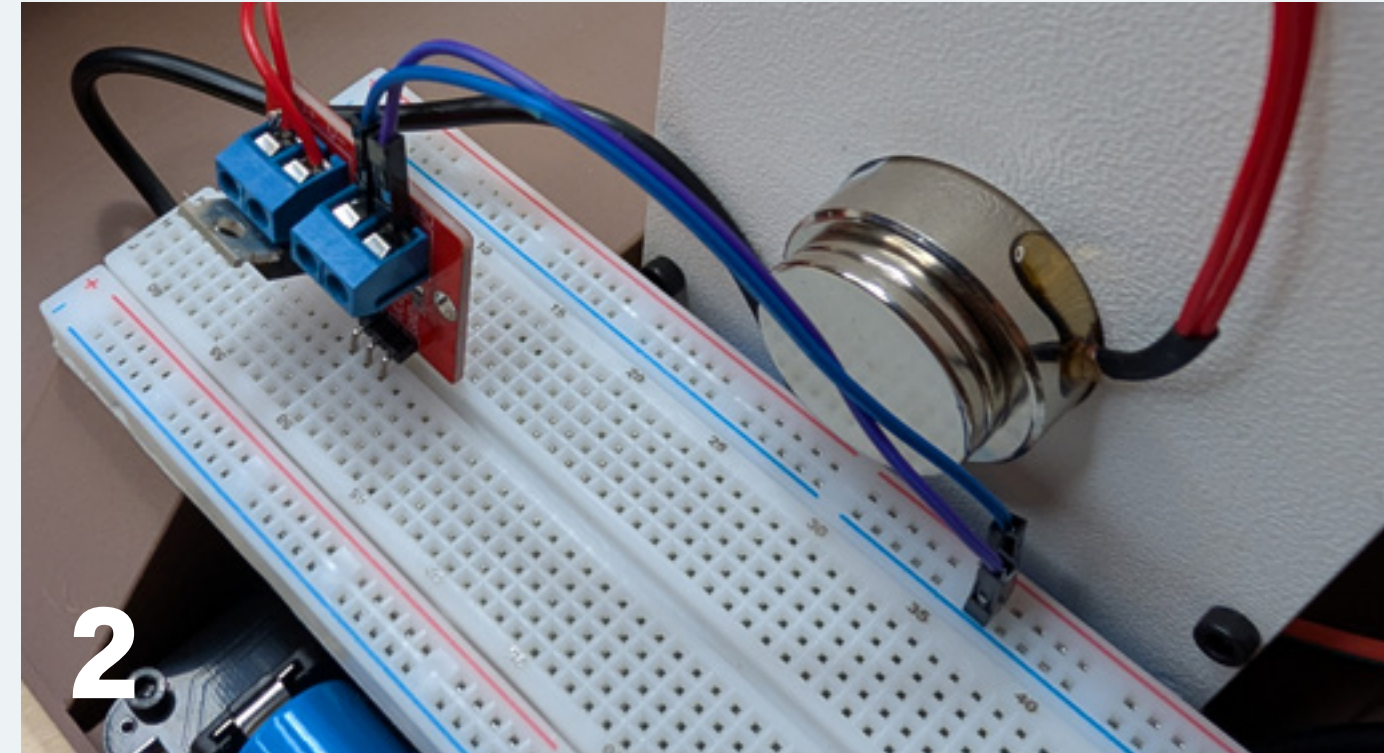
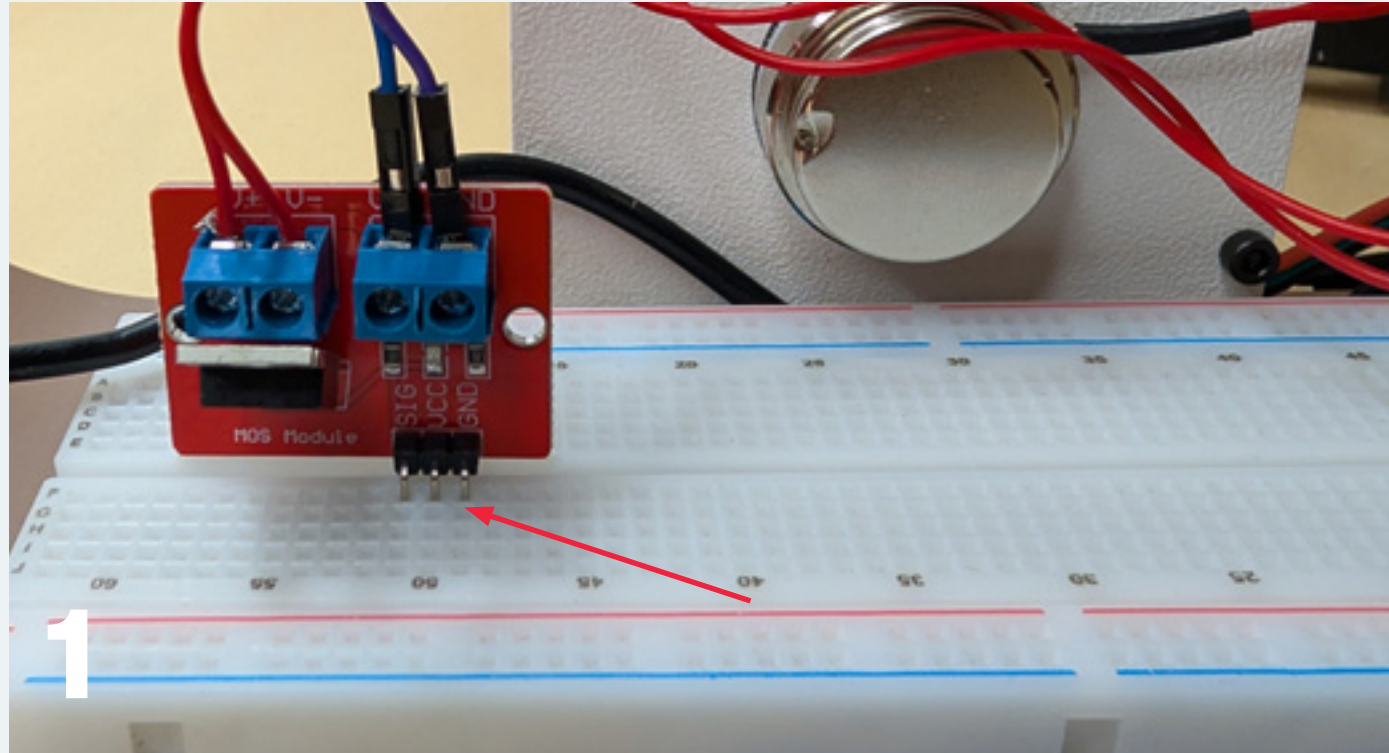


2

ASSEMBLY

2.2

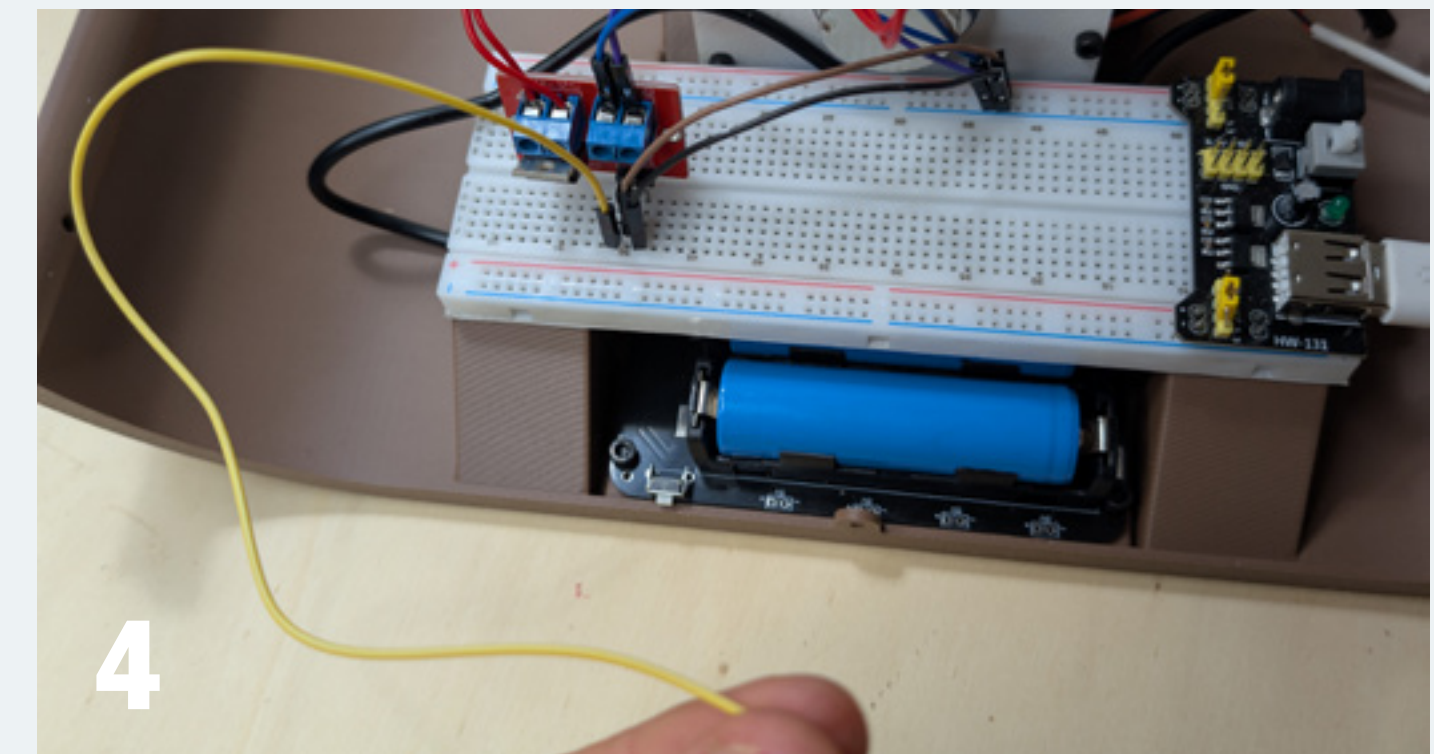
MOSFET TRIGGER and ELECTROMAGNET



Place the MOSFET onto the Breadboard.

First, the two loose Dupont cables we wired in the previous step connect to the 5V power rail we created. We do the same with the MOSFET module VCC and GND, connect it to the 5V rail.

Lastly, we take a Dupont male to female cable (a bit longer) and connect it to the SIG of the MOSFET module. Leave the other end disconnected for now.

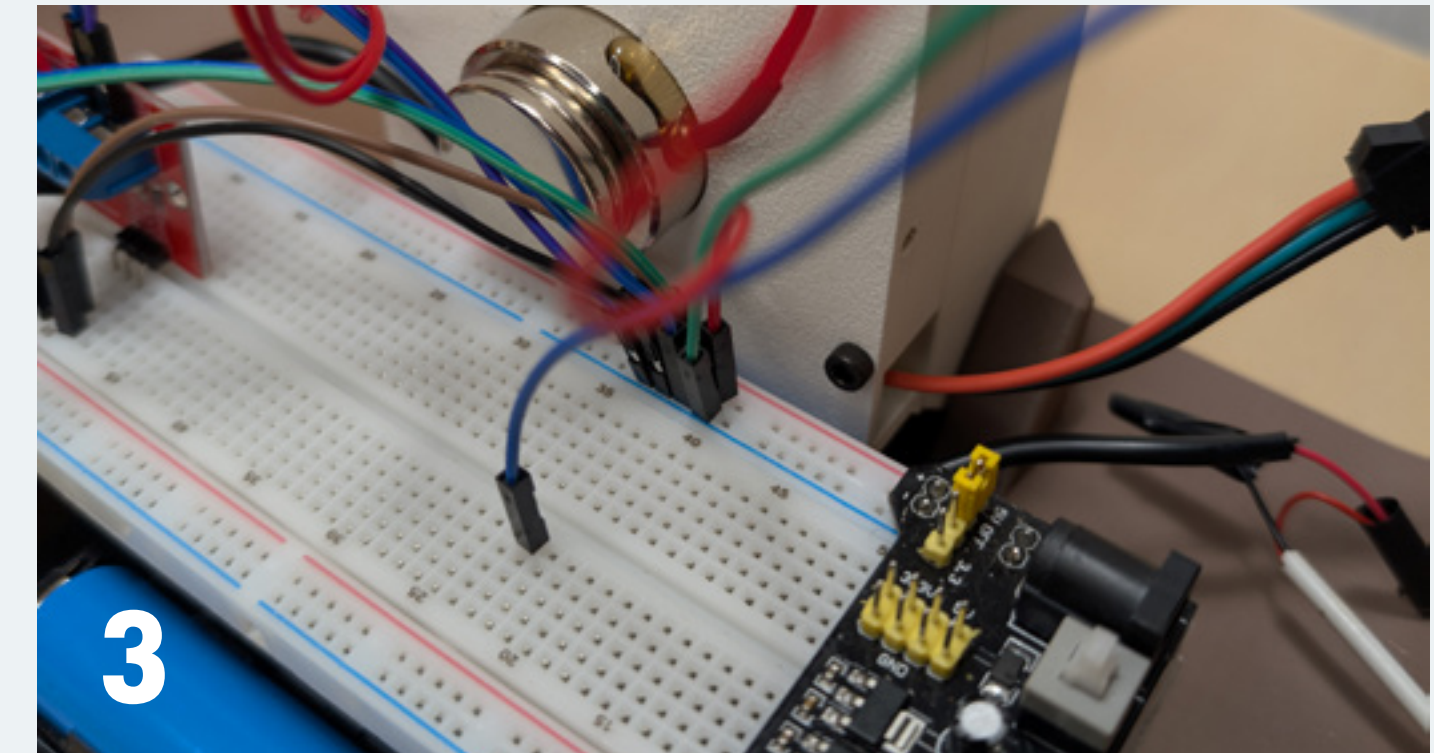
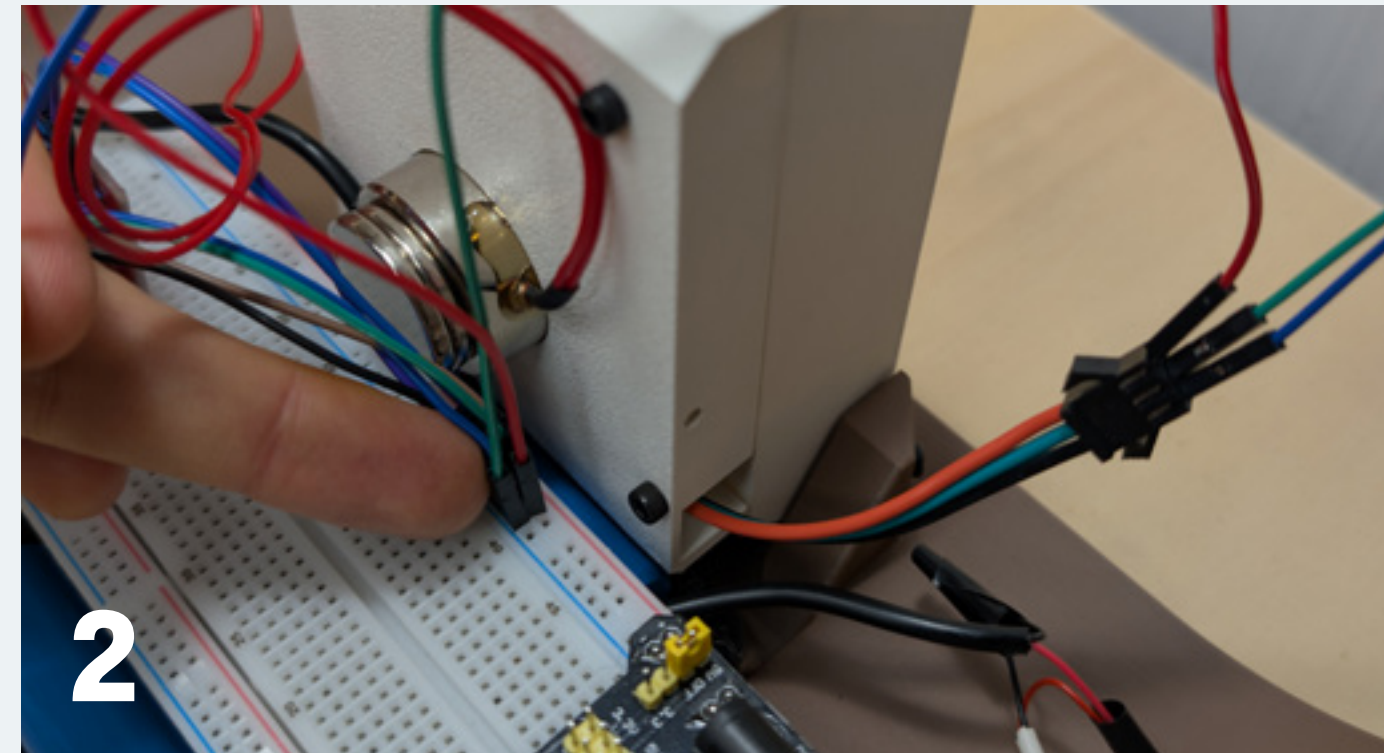
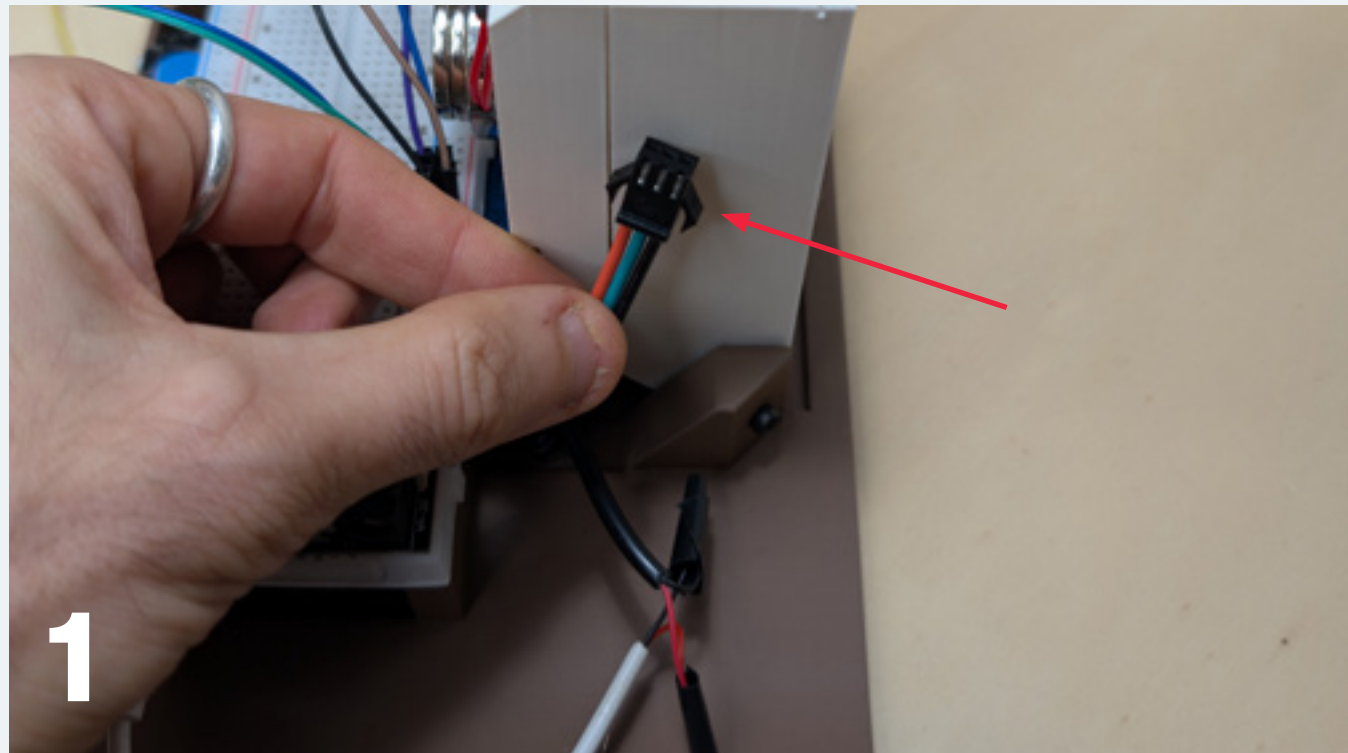


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ASSEMBLY

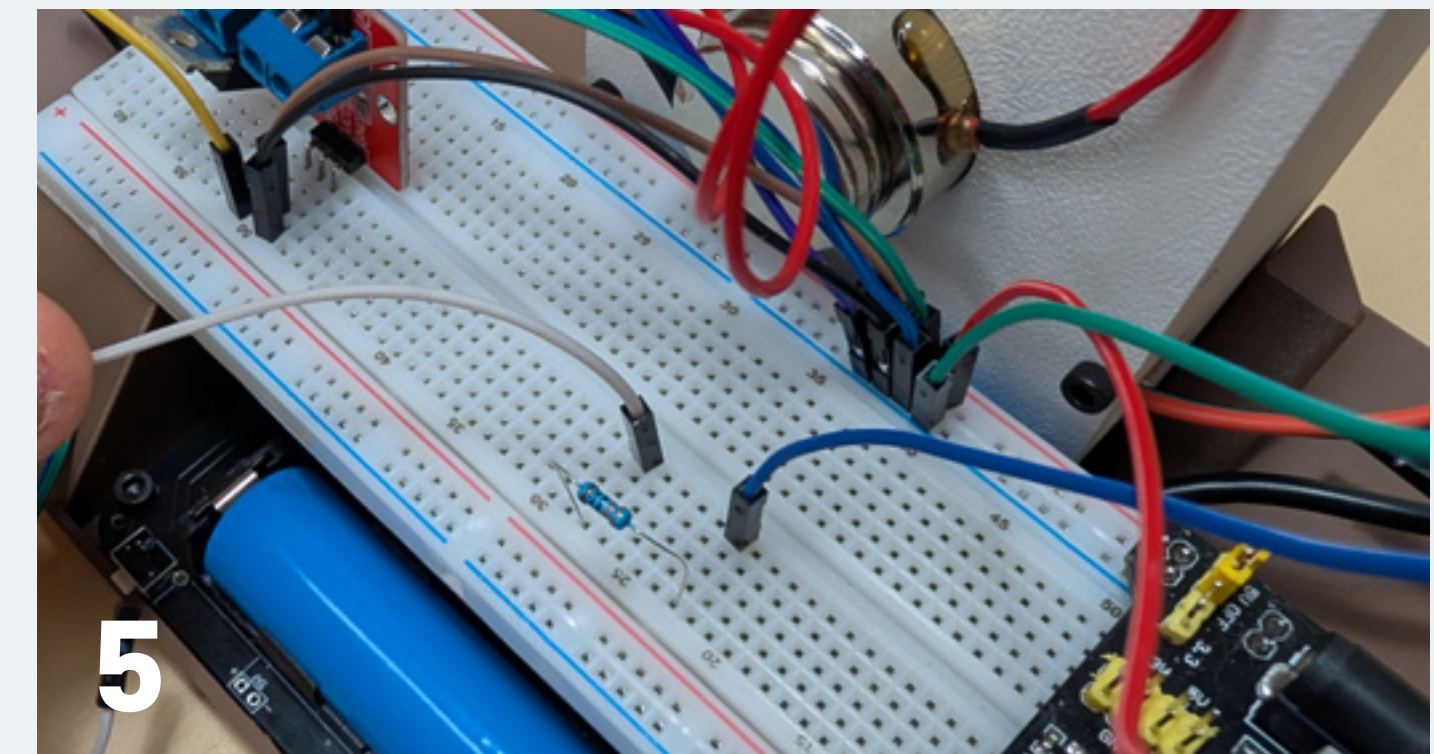
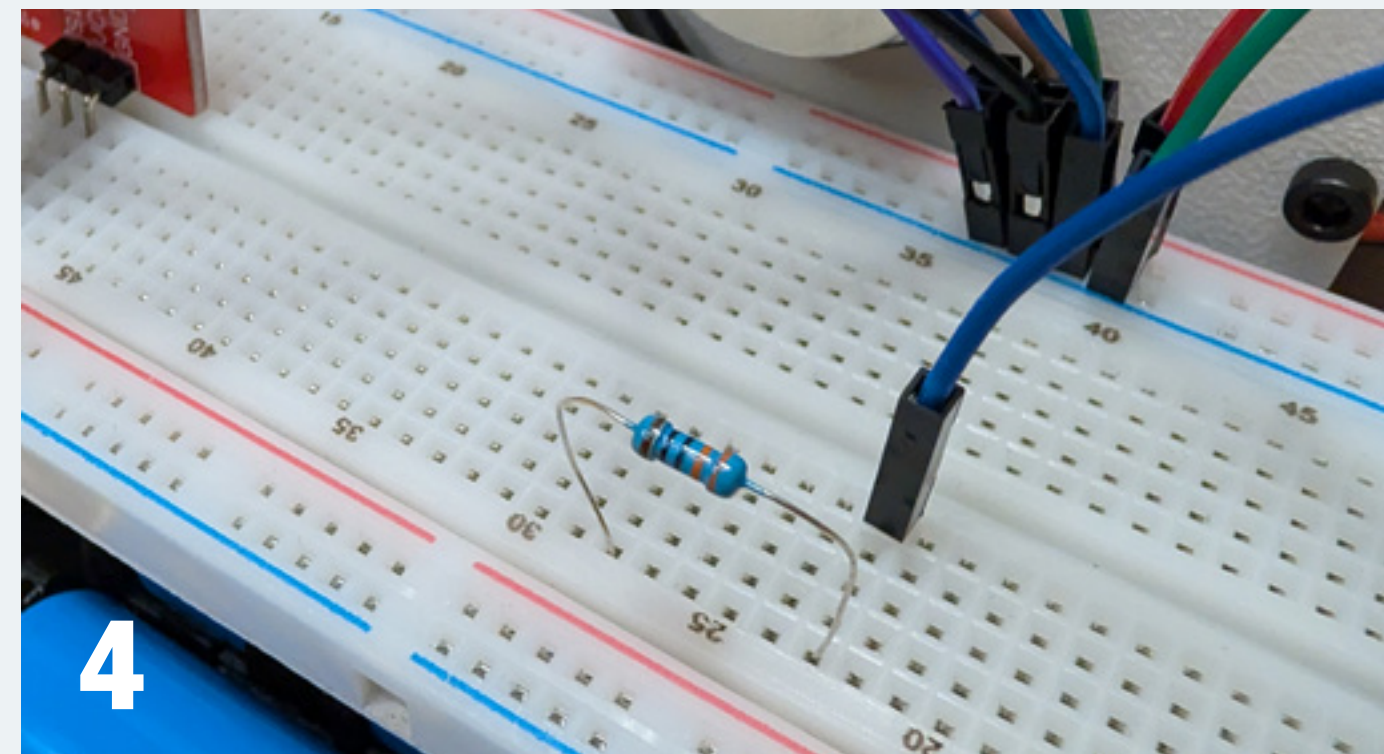
2.2

LED RING



Take the LED Ring cable. It's Vin and GND should connect to our 5V rail.

For the SIG wire, we place it on a breadboard, add a 330 Ohm resistor, then connect a Dupont male-female cable on the other side and leave it hanging for now.

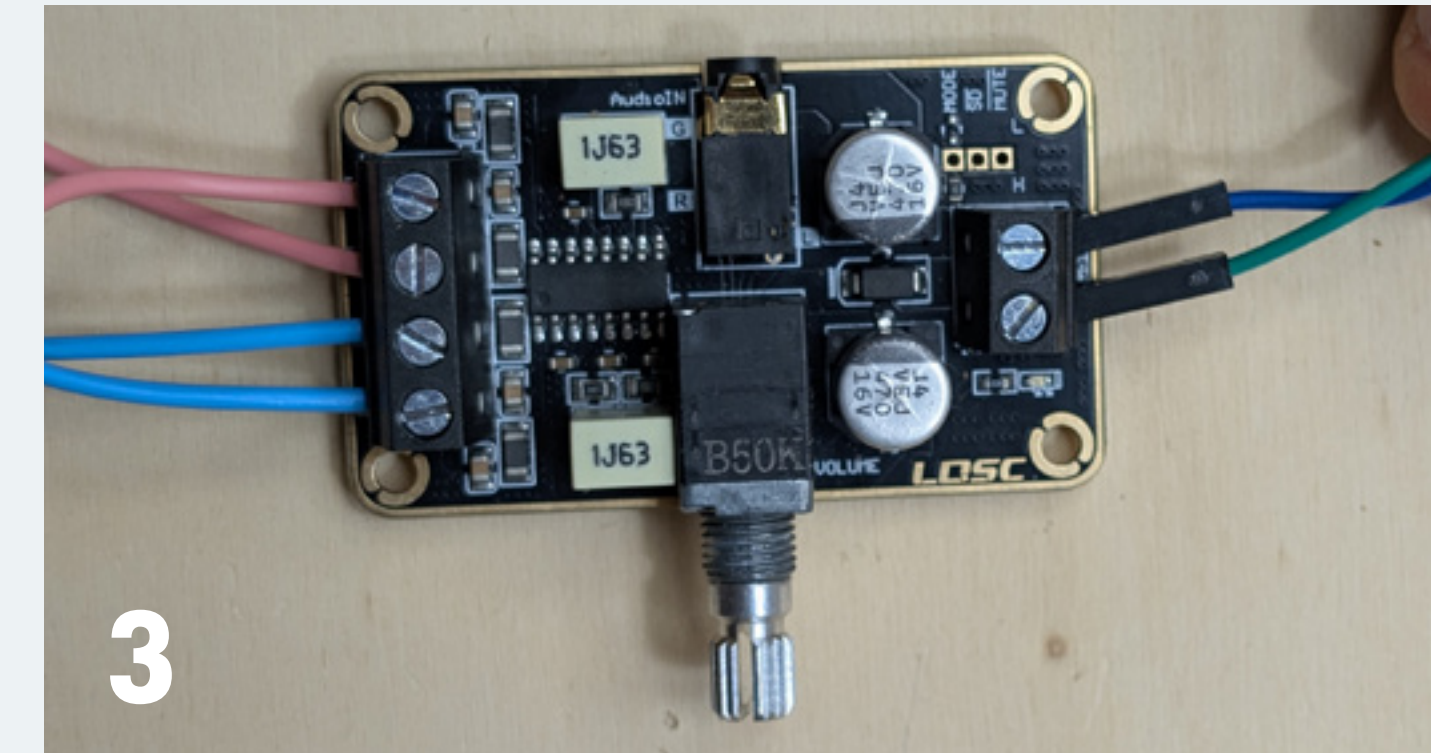


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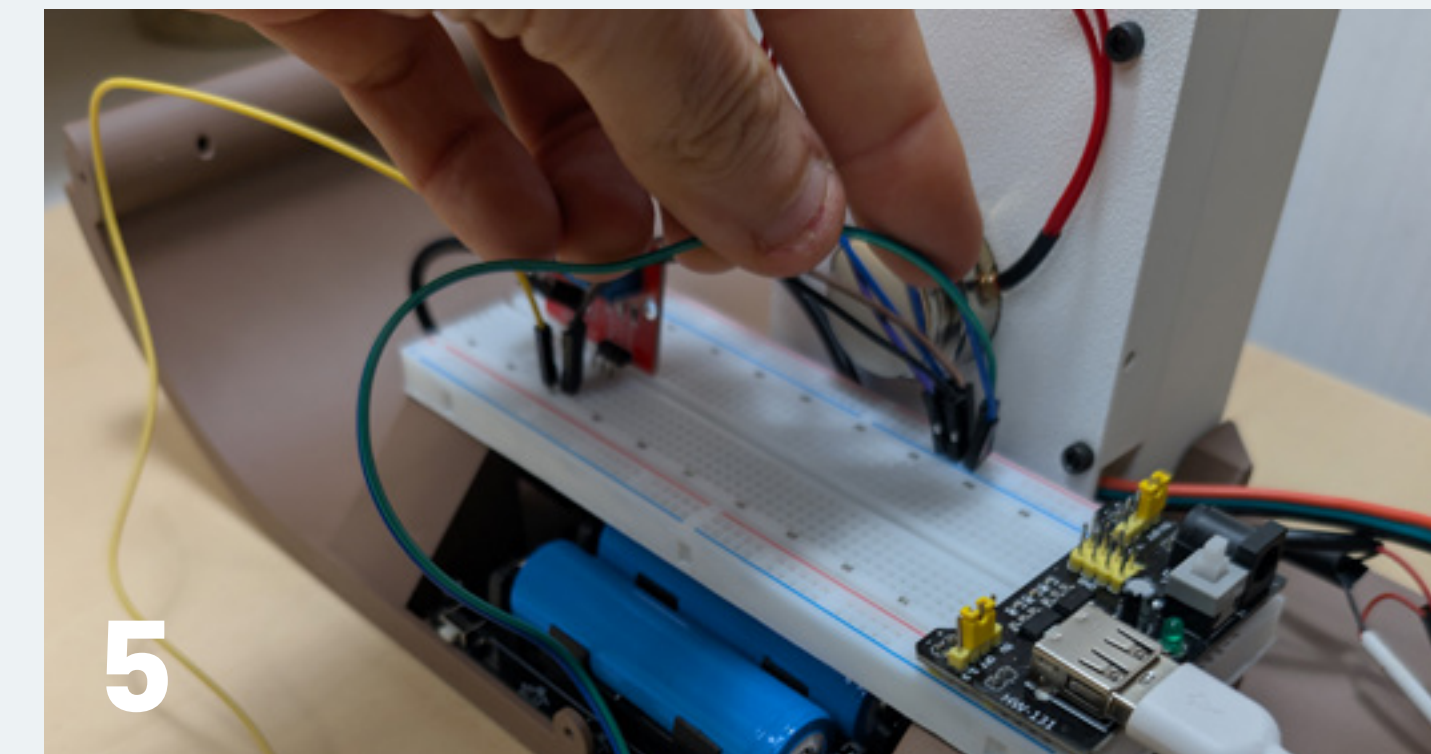
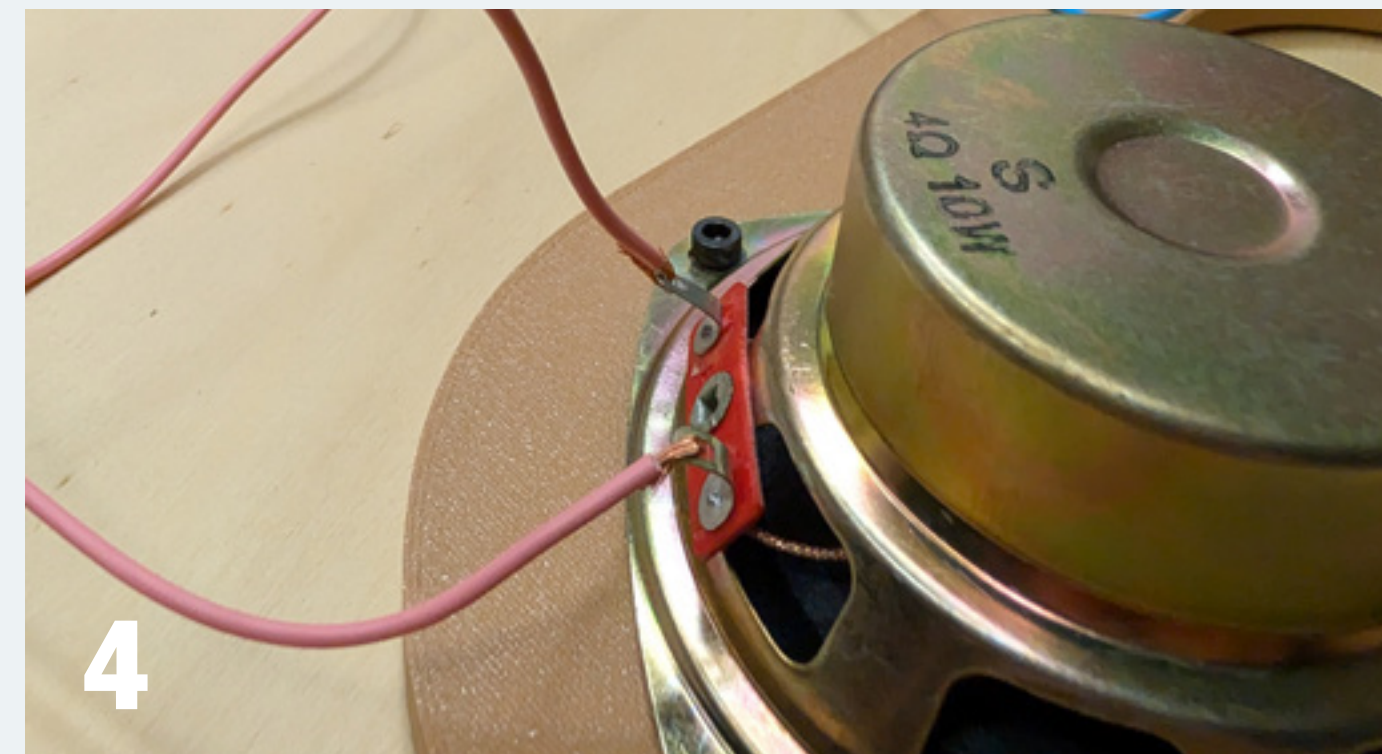
2.2

AUDIO AMPLIFIER and SPEAKERS



Screw the speakers onto the back of the Front Panel with 4 M3x3mm screws.

Take the Audio Amp board. On one side connect the L and R speakers, on the other connect the Vin and GND to our common 5V rail.

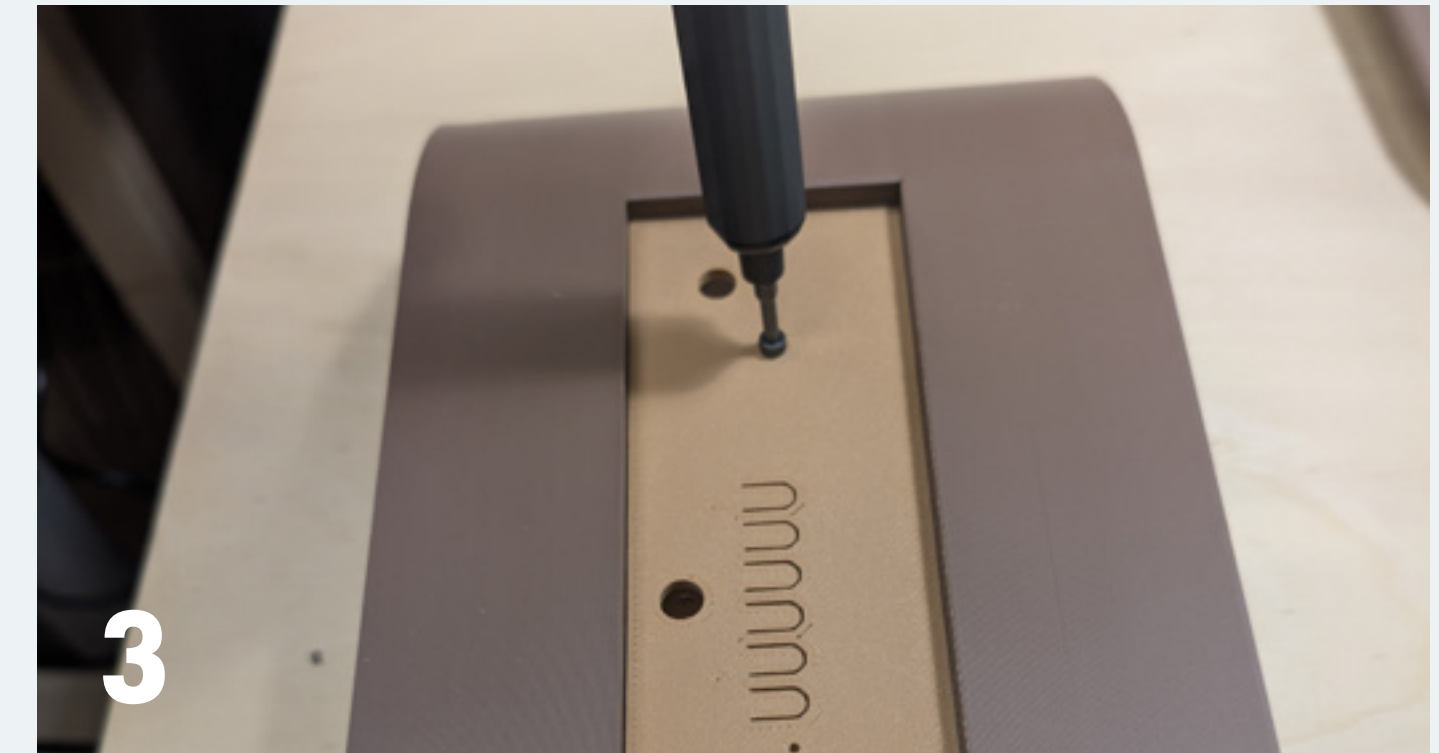


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ASSEMBLY

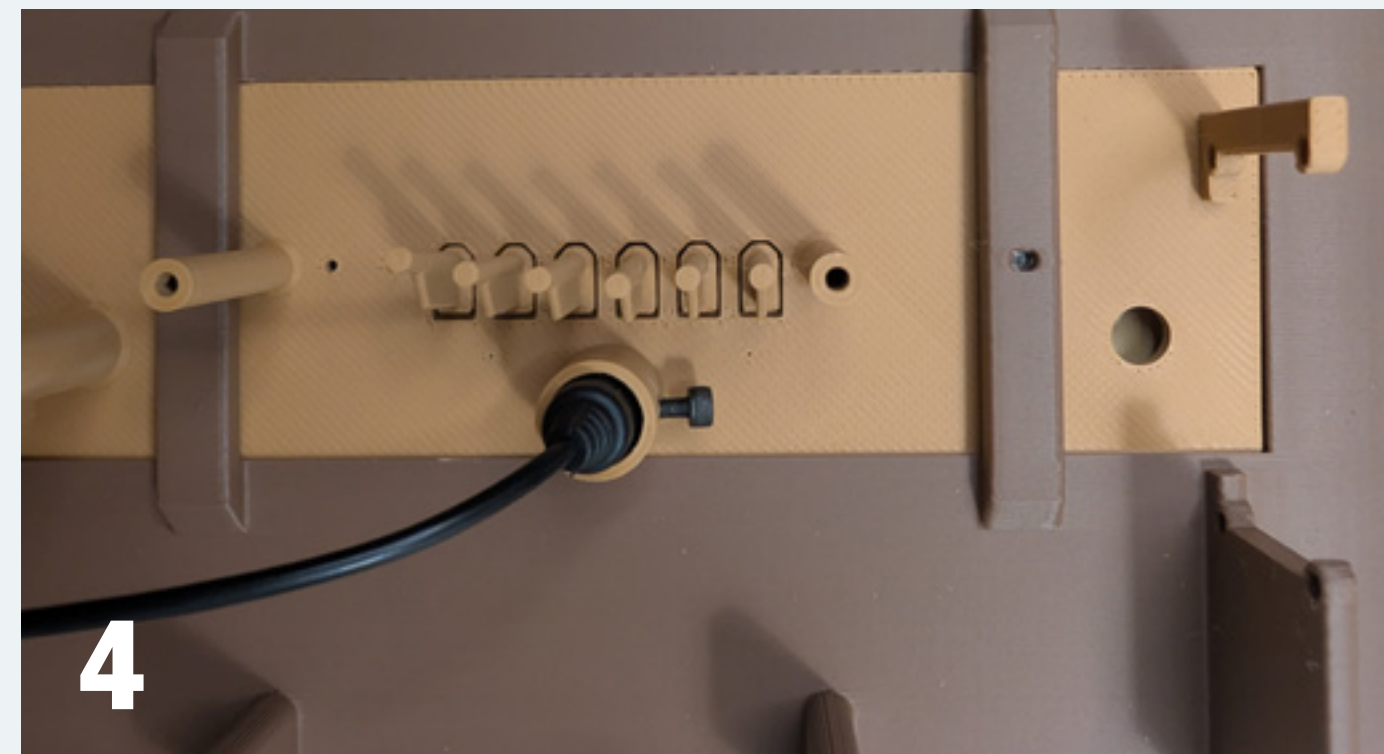
2.2

TOP PANEL



Take the Top Panel and Control Panel and join them together. If unsure which side is up, take a look at image 2. From the top, screw them together via two M3x6mm screws.

Take the short female-male mini jack cable and secure it in place.

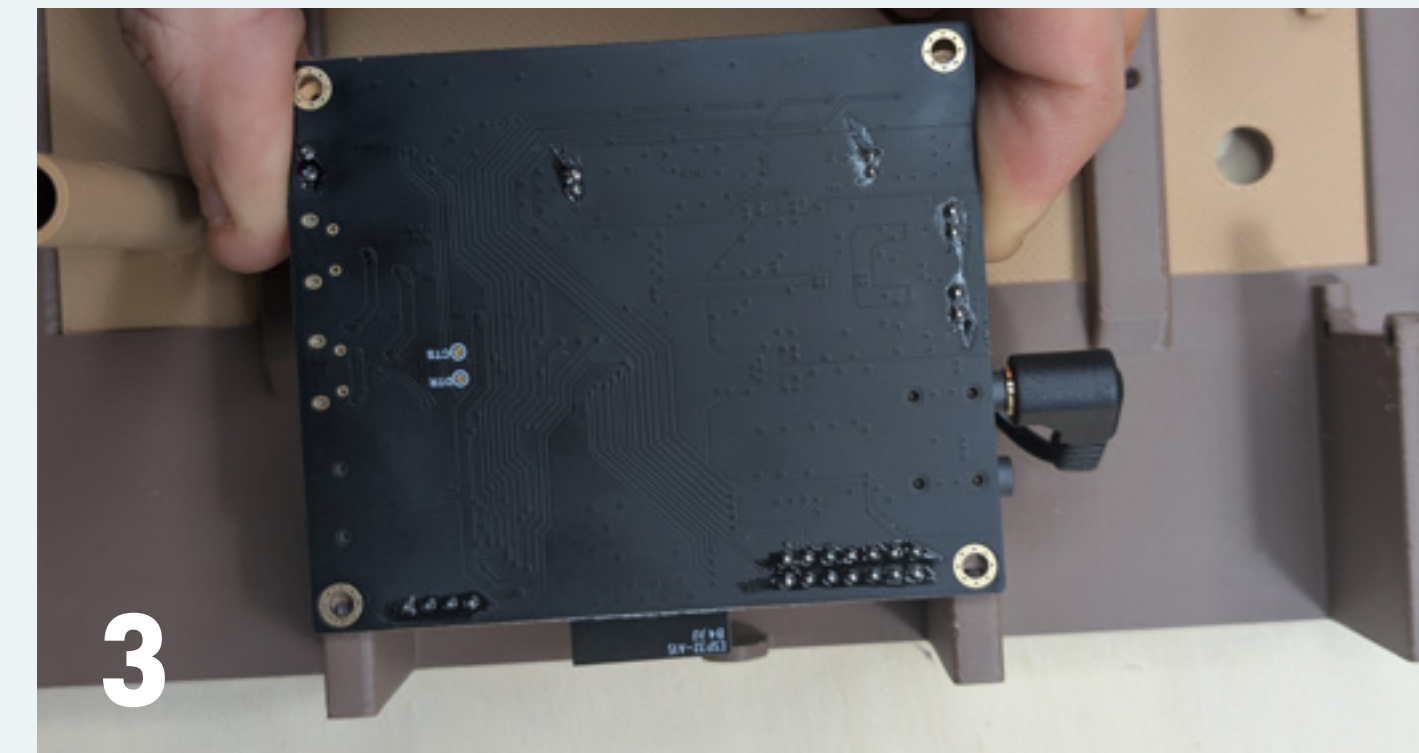
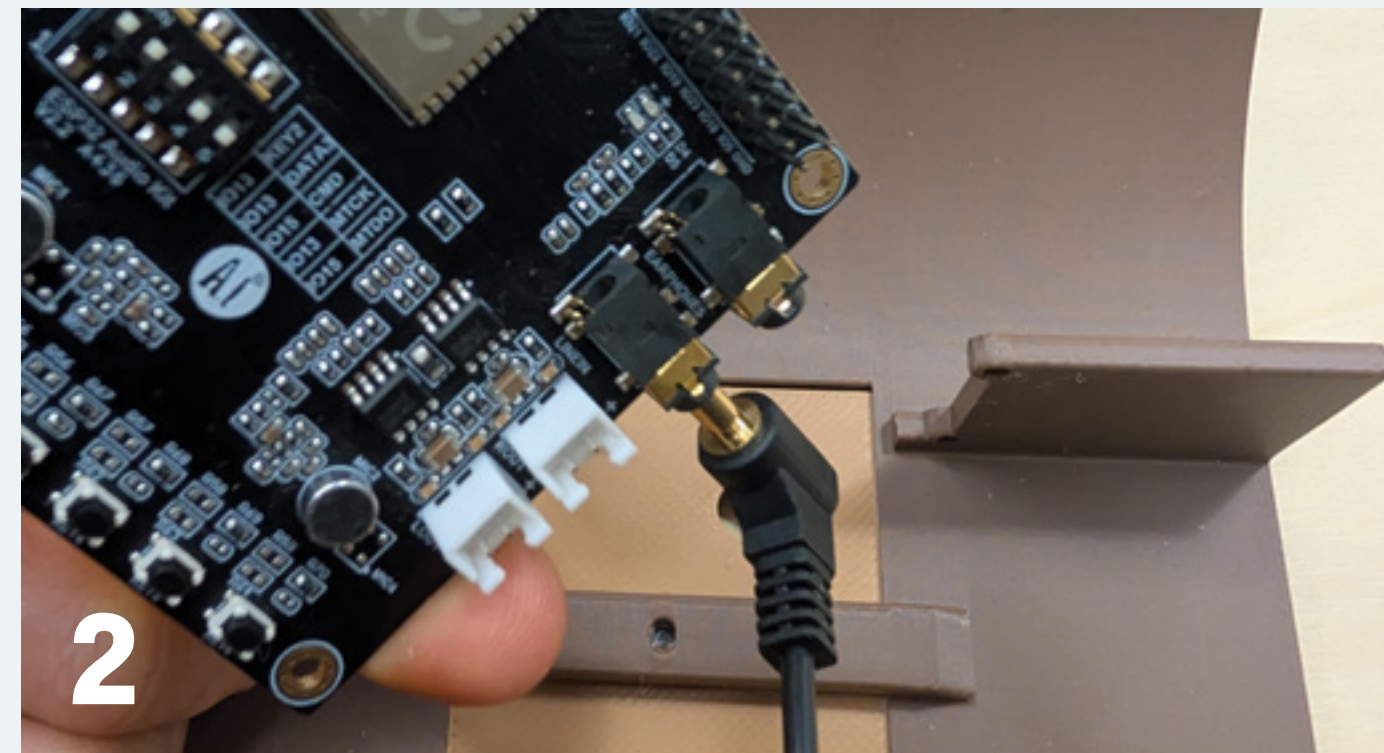


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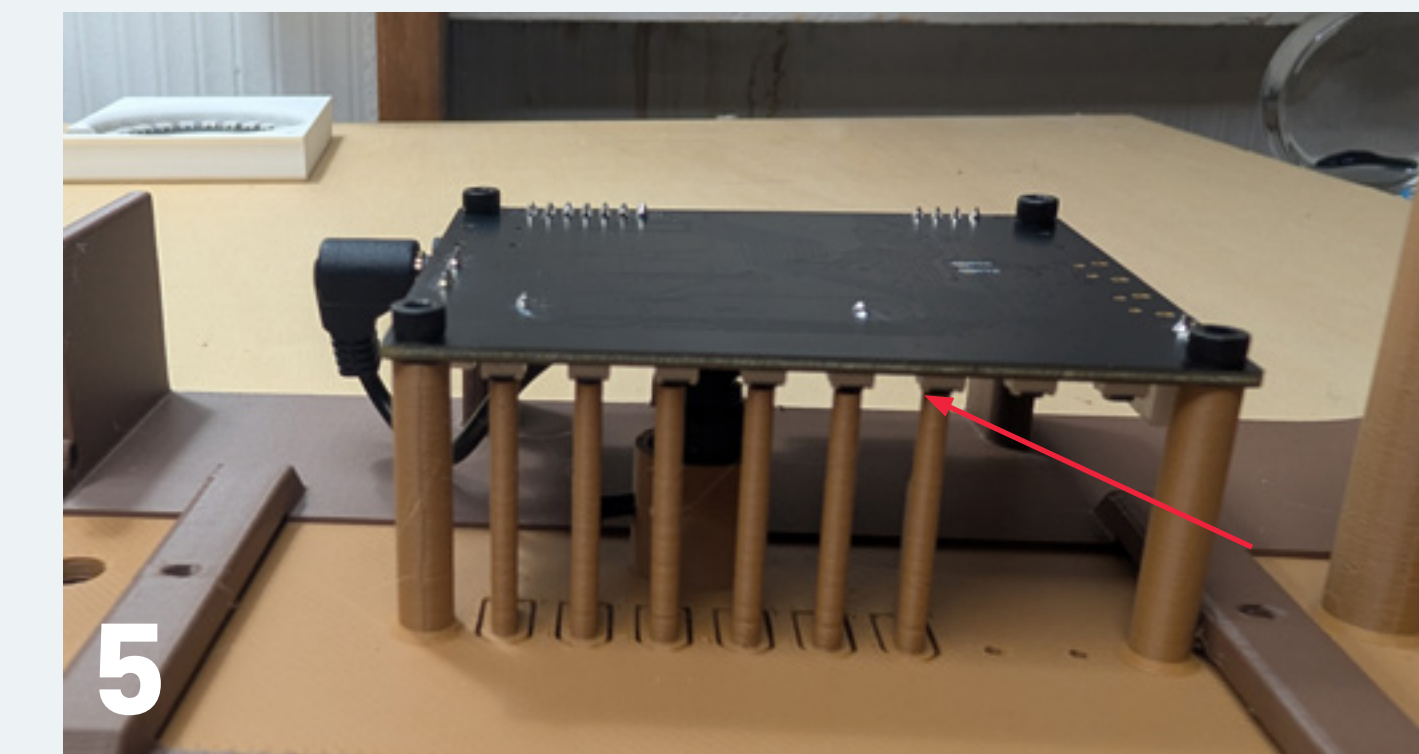
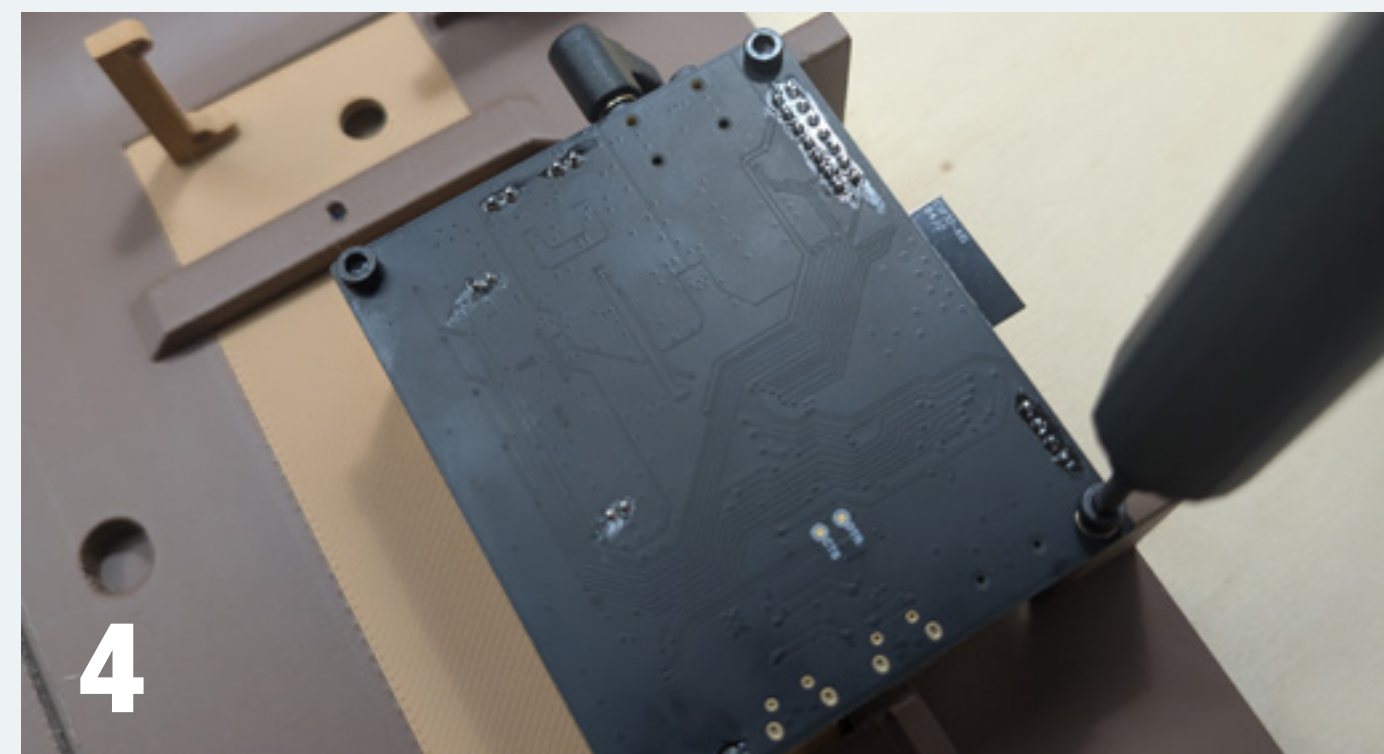
2.2

AUDIO BOARD



Take the ESP32-A1S Audio Kit Board, our brains. Connect the jack cable we just secured in place and connect it to LINE IN.

Place it on top and screw it in with four M3x3mm screws. Watch out for the buttons (Image 5) - they should align without clicking the button.

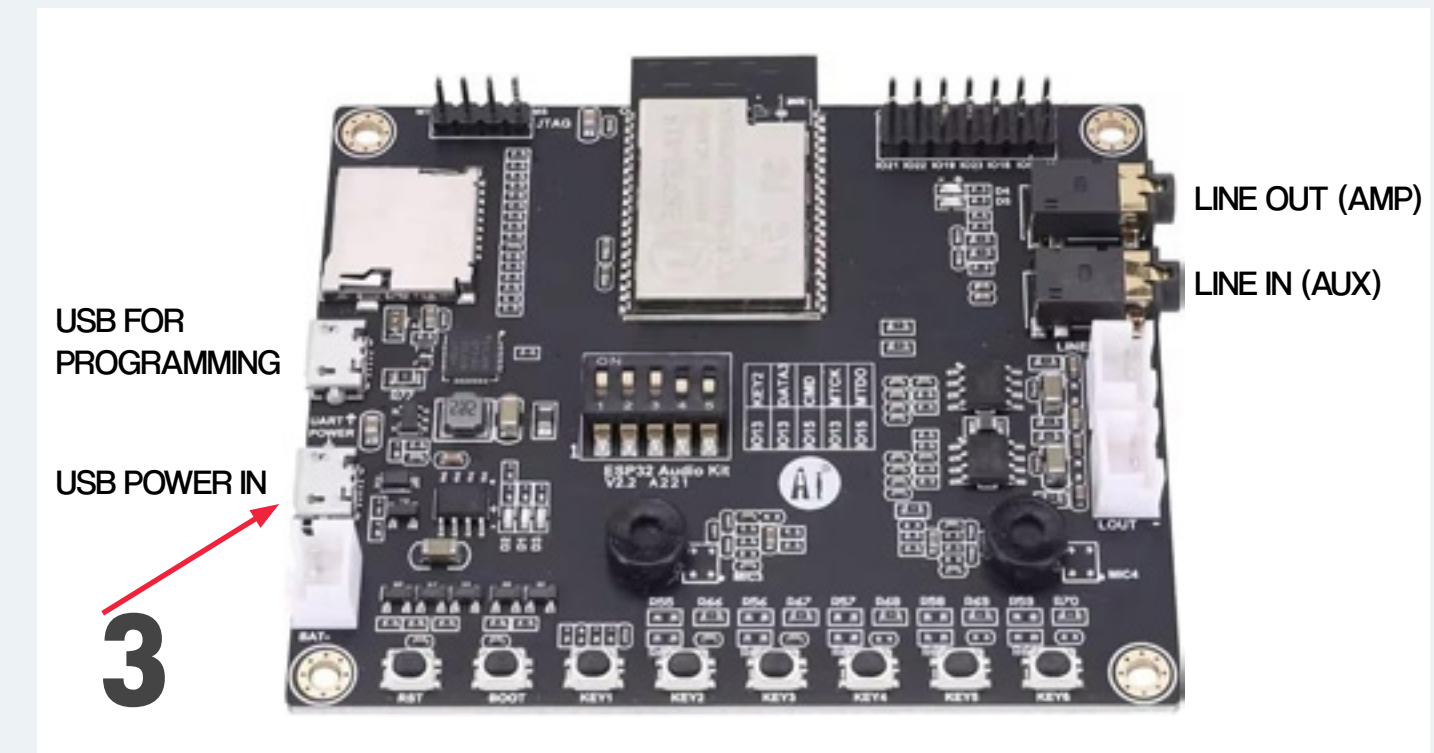
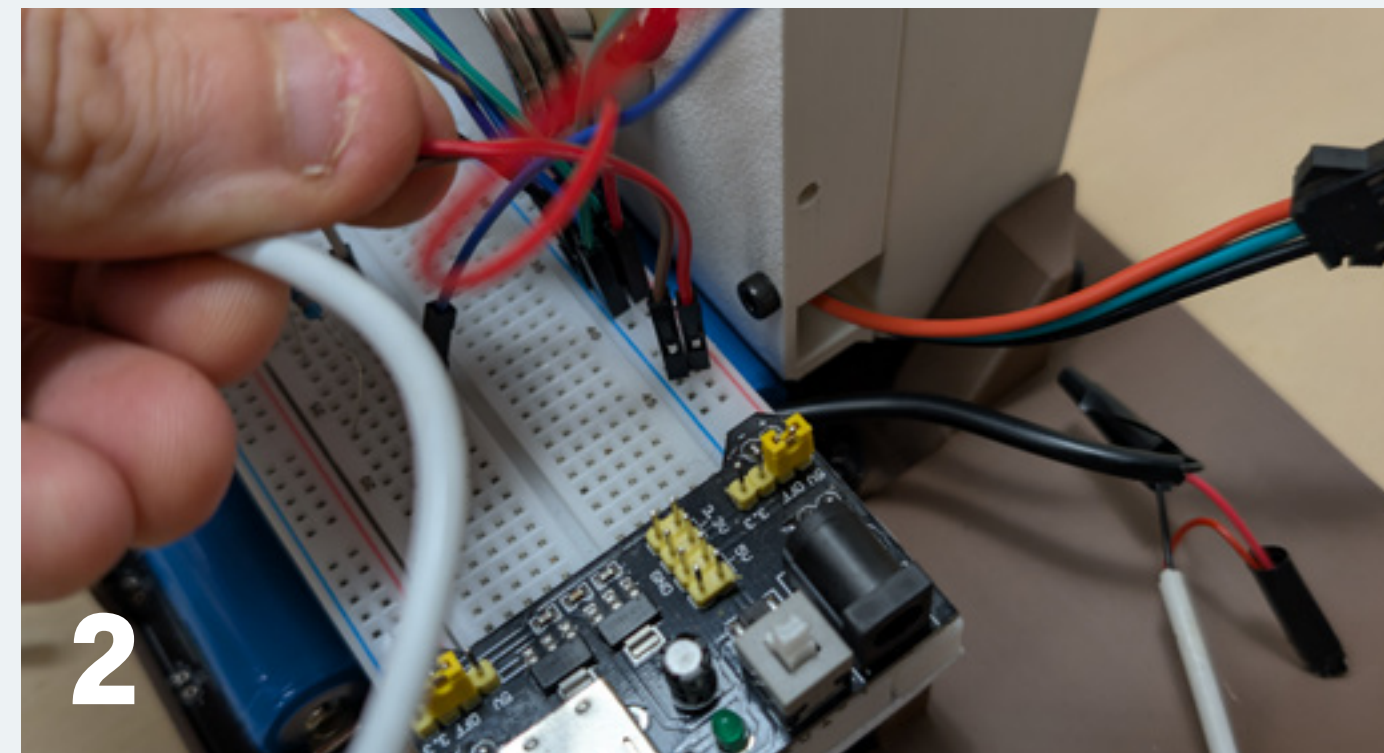


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ASSEMBLY

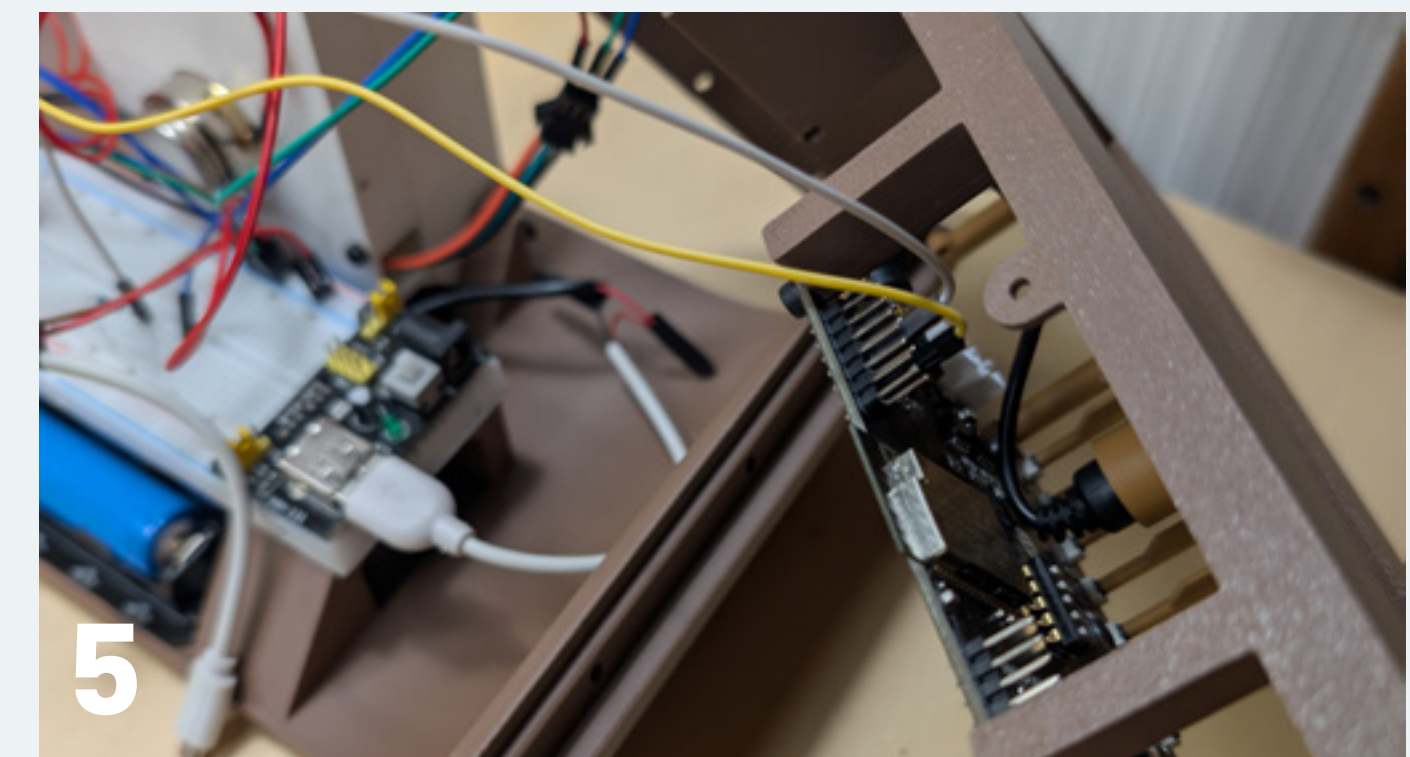
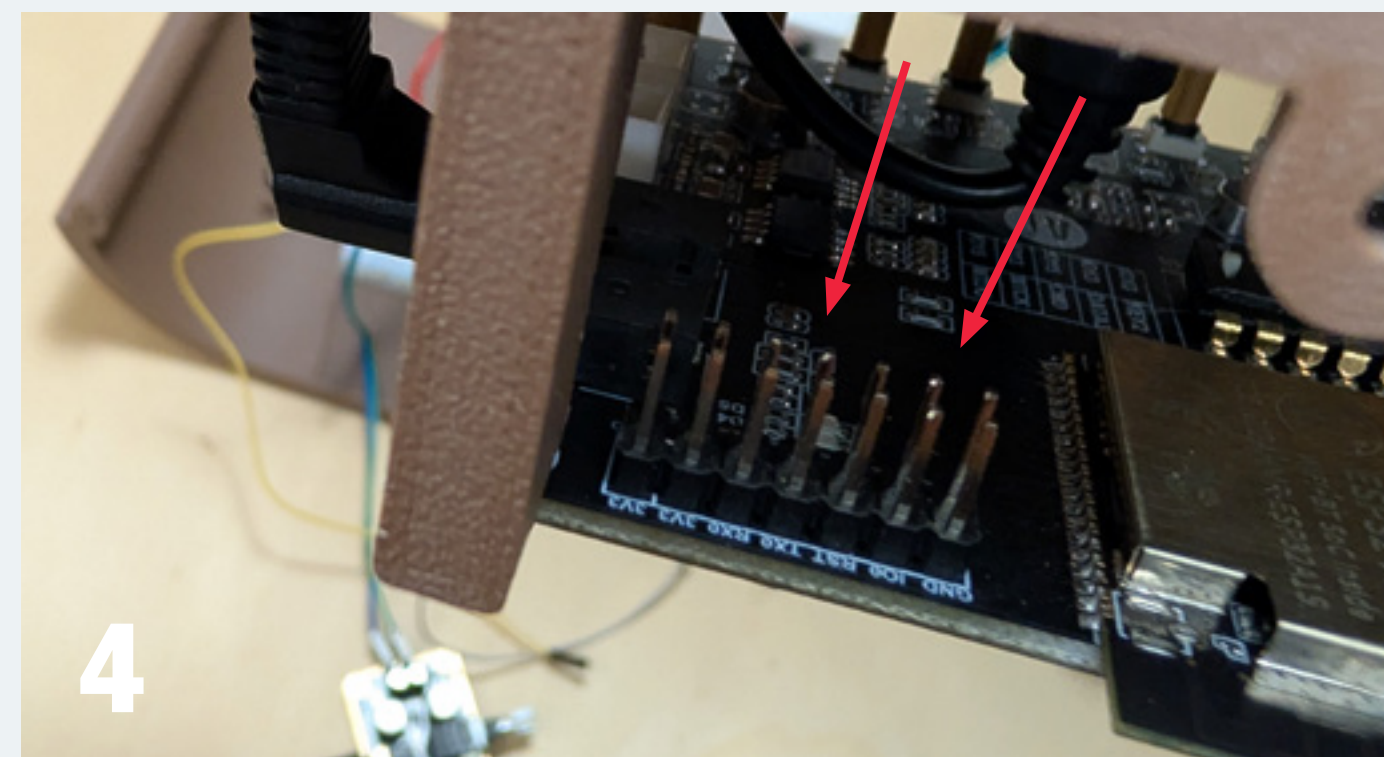
2.2

AUDIO BOARD



We need to create another cable - from male Duponts to Micro-USB. The Duponts go to our main 5V power rail, the Micro-USB part goes to the USB-power IN on the ESP32 Audio Kit Board.

From our previous steps we have a loose MOSFET trigger signal cable and LED Ring signal cable. These go to IO22 and IO23.

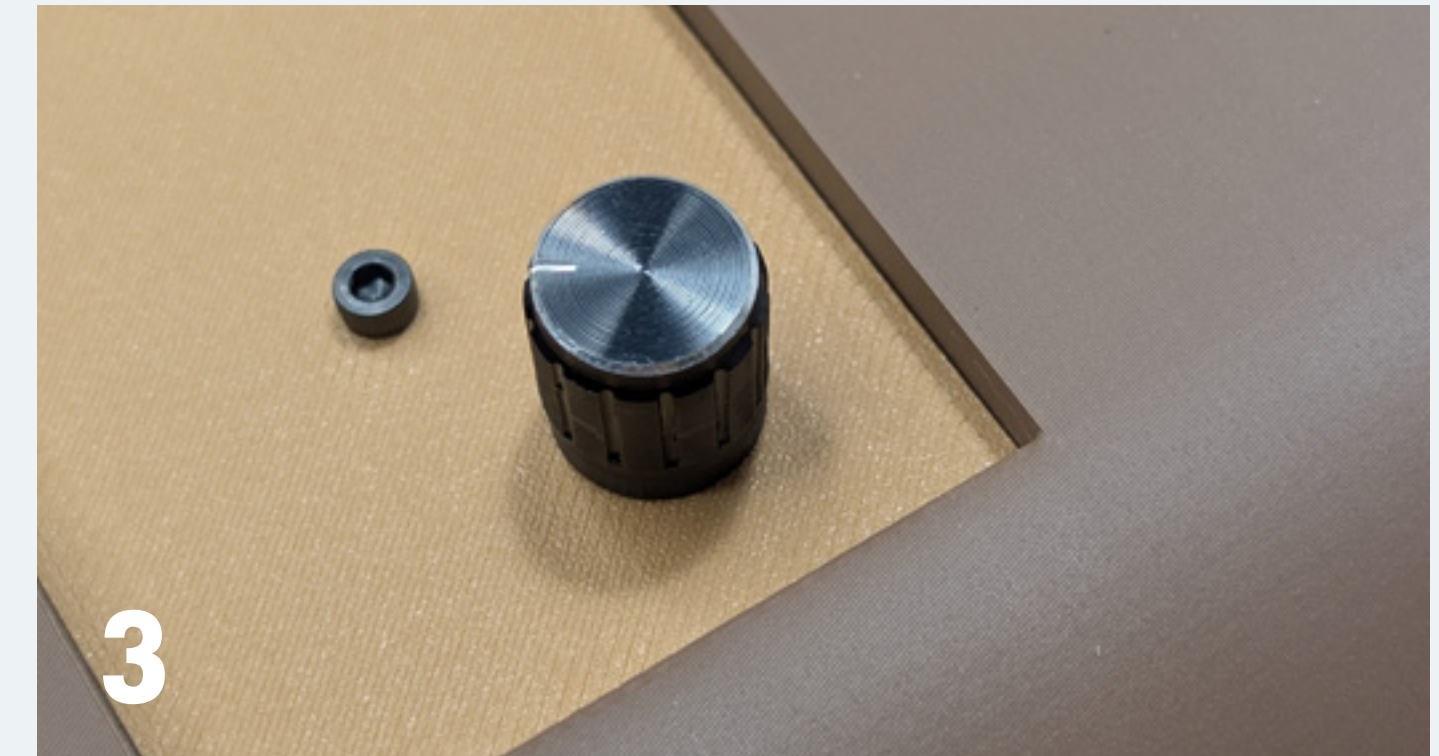
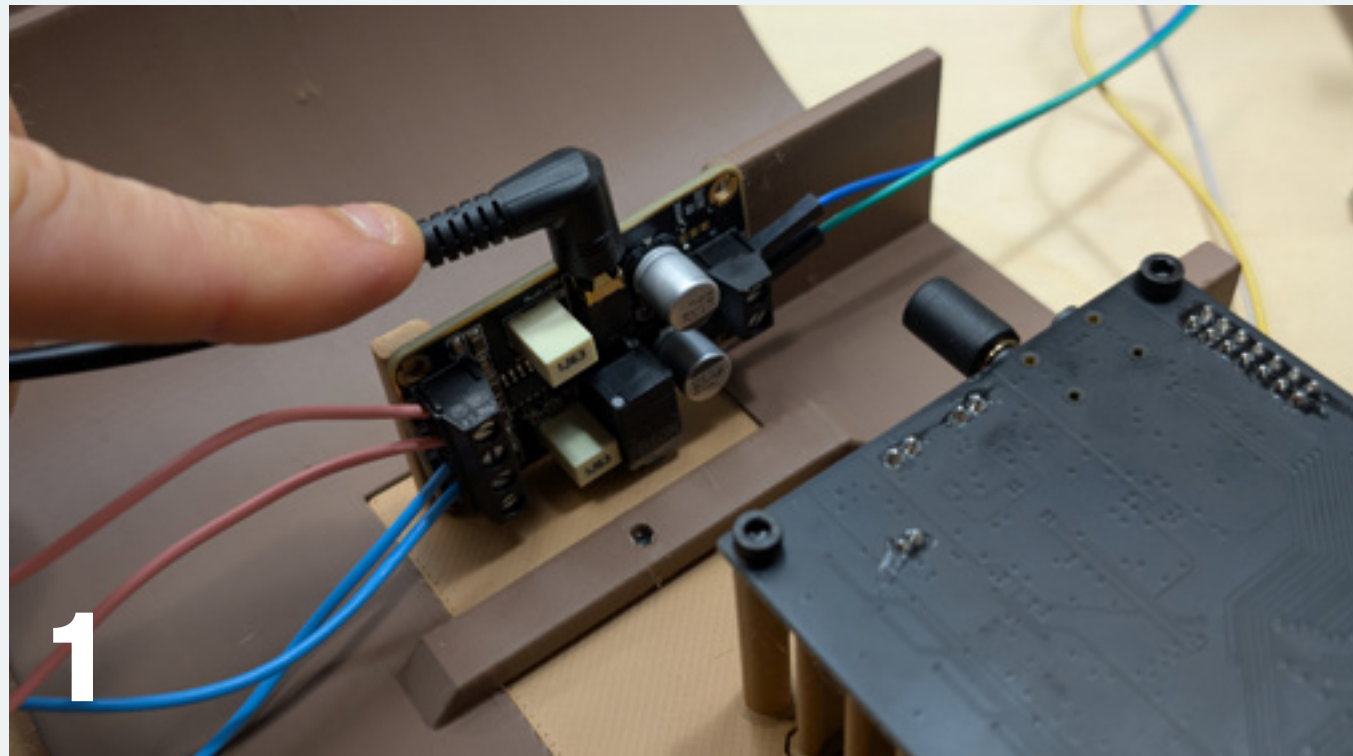


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ASSEMBLY

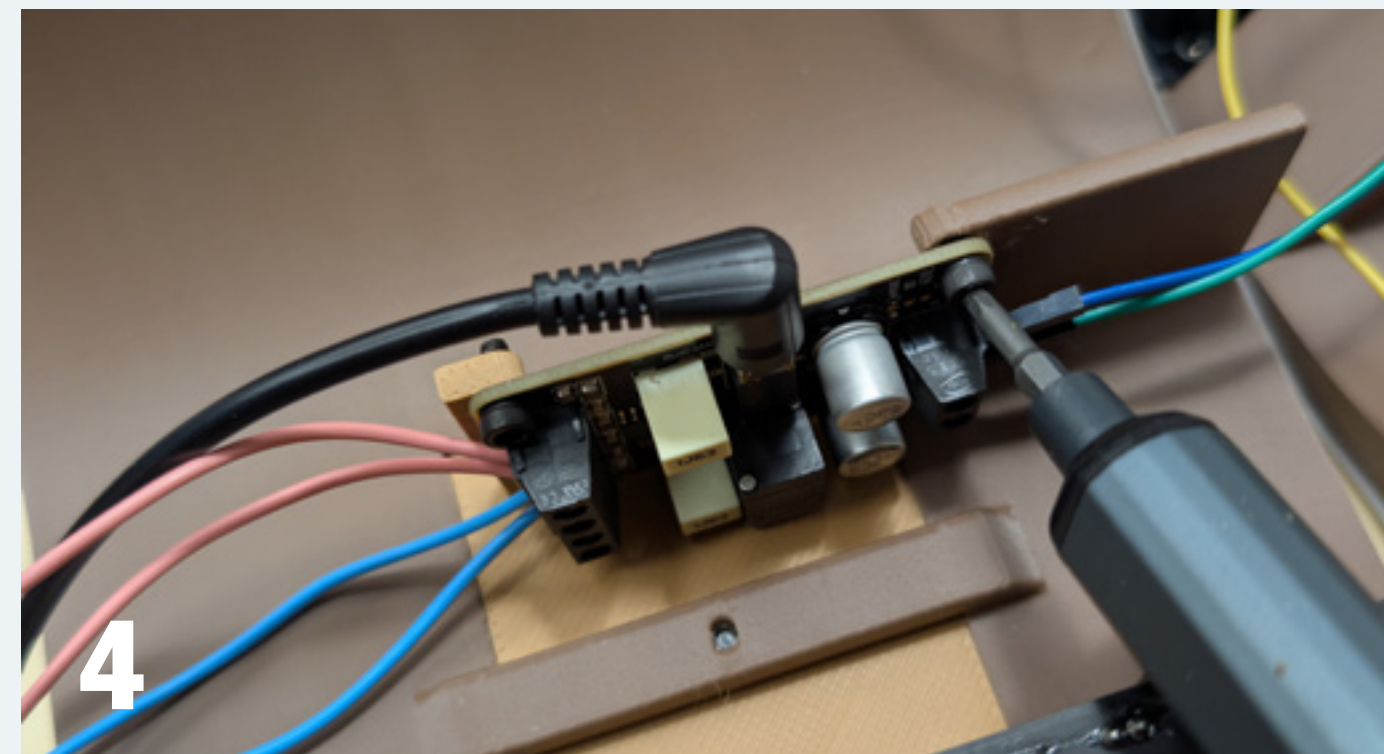
2.2

AUDIO AMP



Take the Audio Amplifier, connect a mini jack - mini jack cable to it. Put it in place, so you can fix it on the other side. Put the cap on. If you want, you can also secure it in place via screws on the inner side.

Connect the jack to the Audio boards Line out.

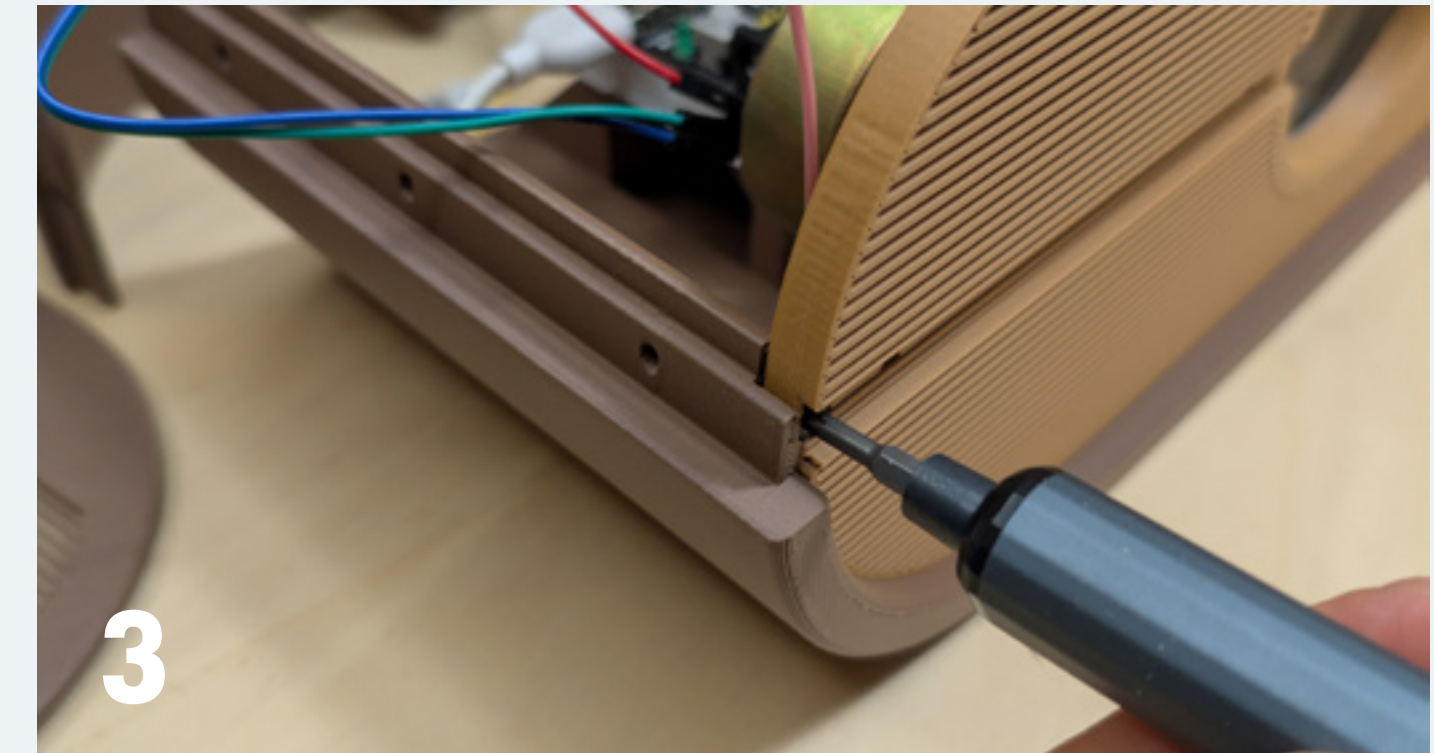
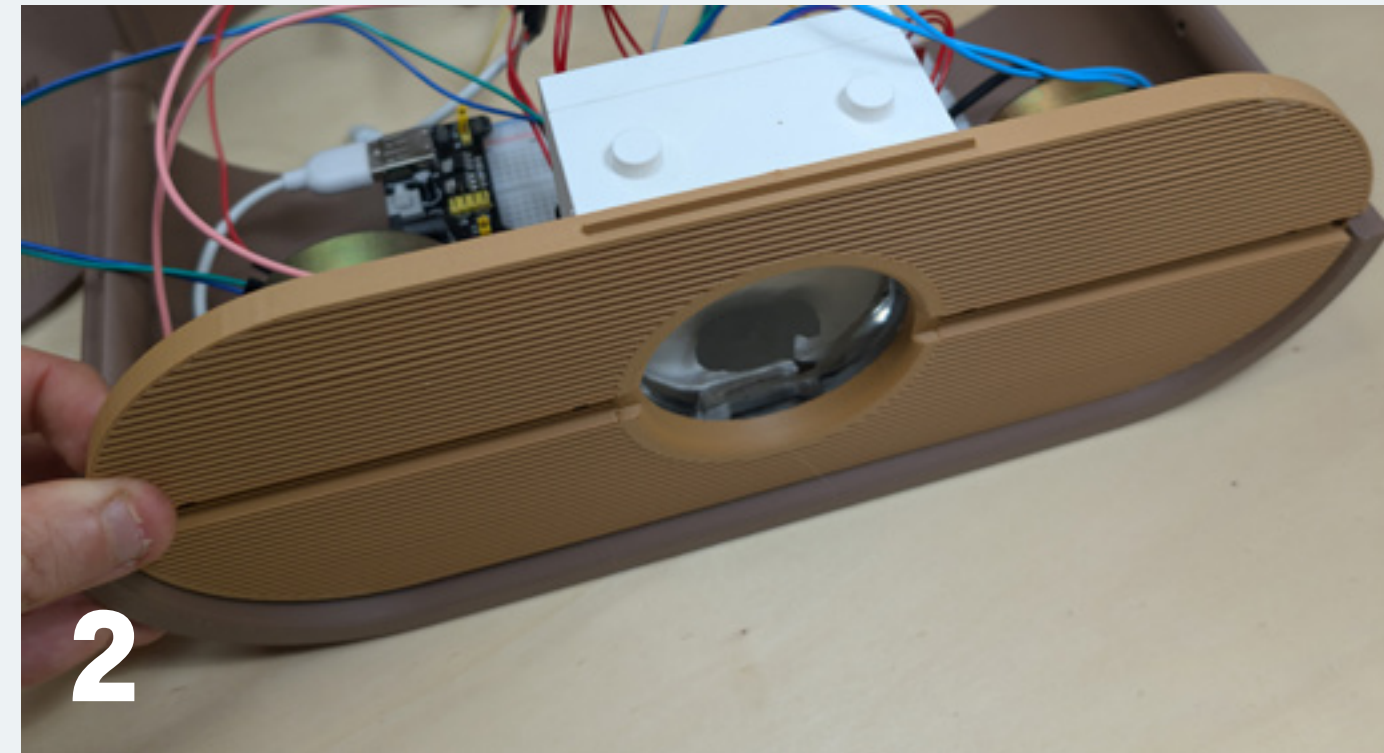
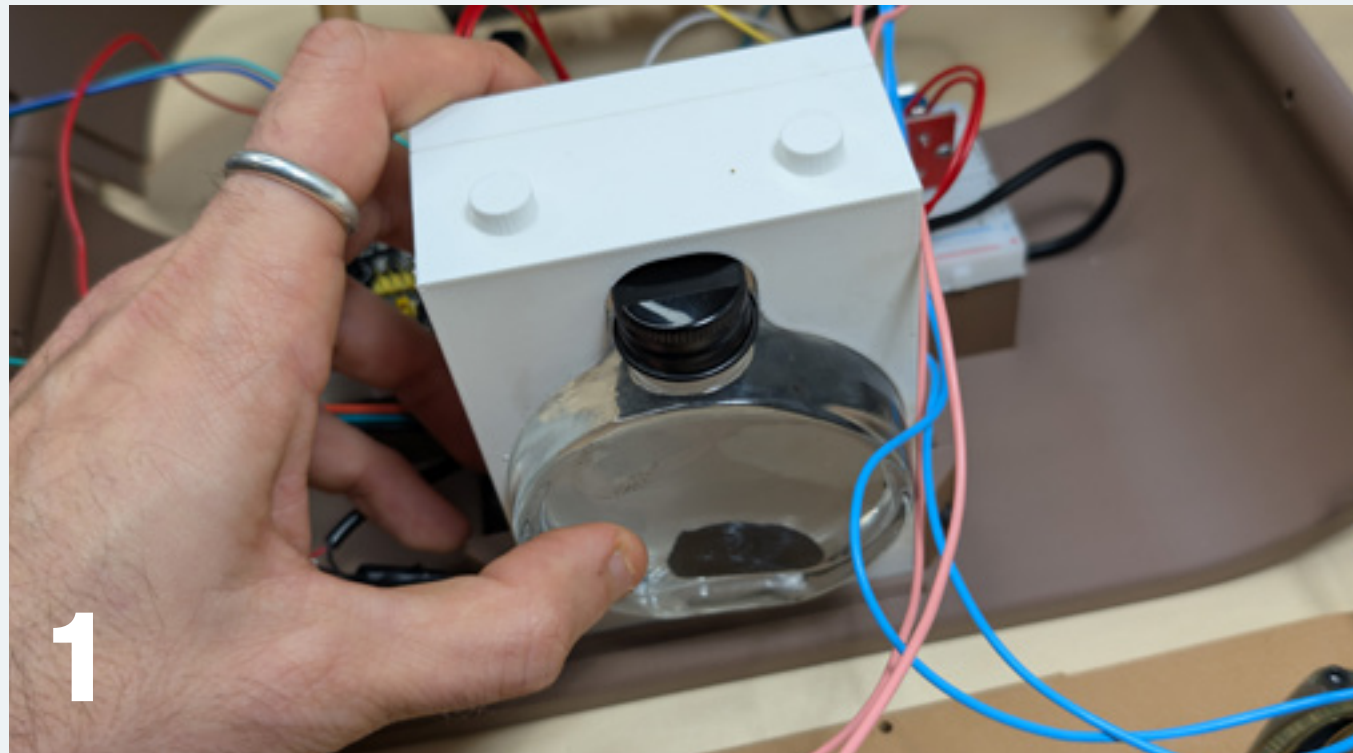


2

ASSEMBLY

2.2

PUTTING IT TOGETHER



Take the ferrofluid bottle and put it in place. If it feels loose, you can secure it with some doublesided tape on the top, sides and bottom.

Take the front panel and secure it in place using 2x M3x6mm.

On the top panel, assemble the power switch from top/bottom. Make sure it rests on the Breadboard Power module on/off switch.

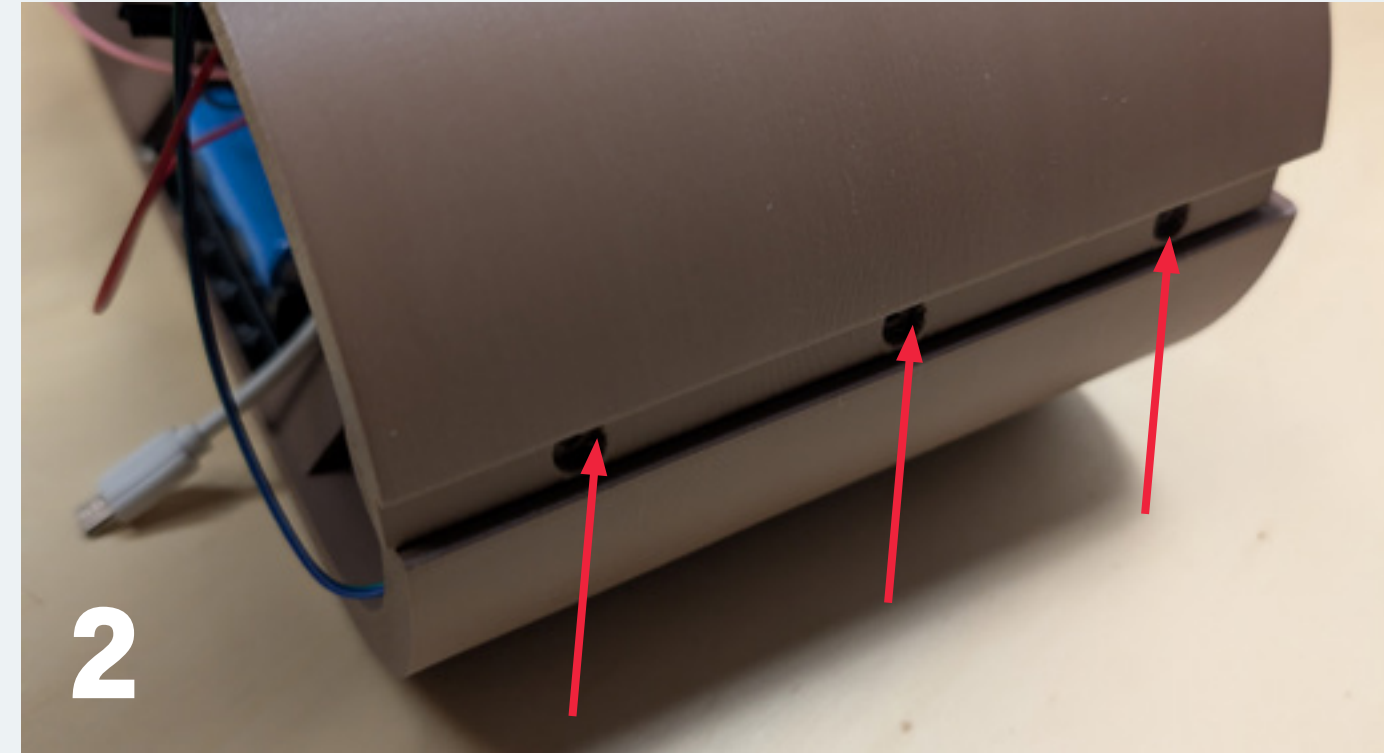


2

ASSEMBLY

2.2

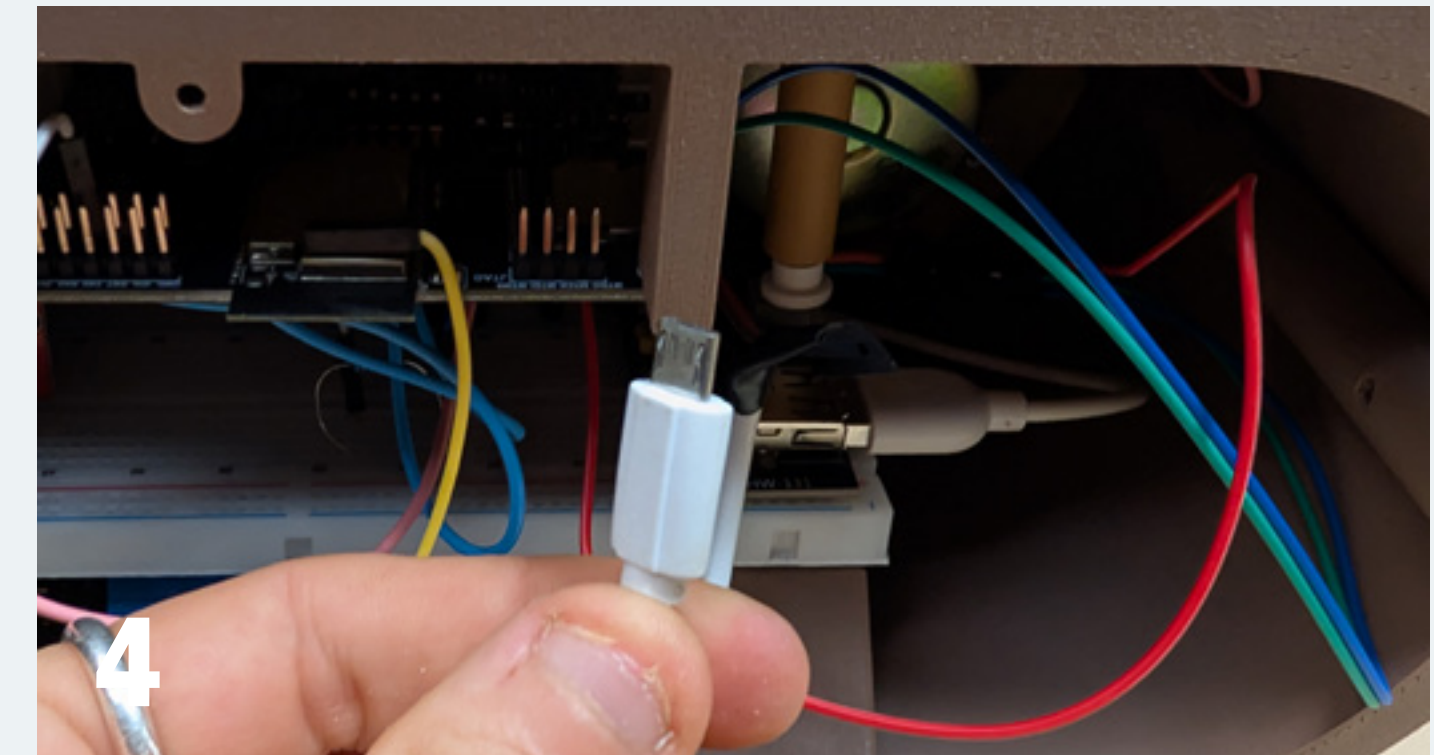
PUTTING IT TOGETHER

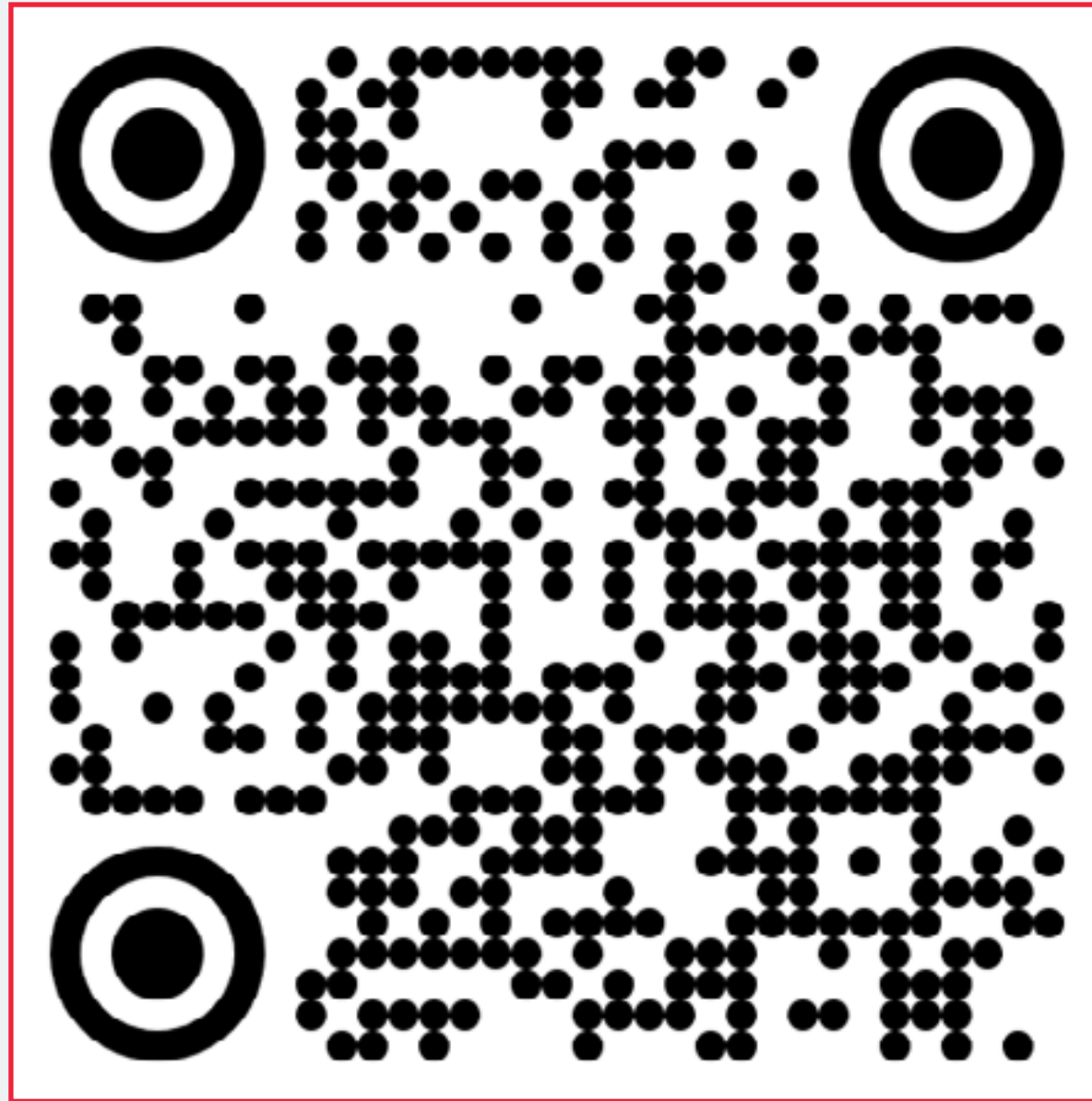


Put the Top Panel on, secure it with three M3x6mm screws on each side.

Slide in the Front Bars, you can also secure them with some superglue.

Take another Micro-USB cable, but this time it has to be a data cable, as we need to program the ESP32 board. Connect the Micro-USB to the UART port on the Audio Kit Board.





You can also find everything on the official FerroWave GitHub page:

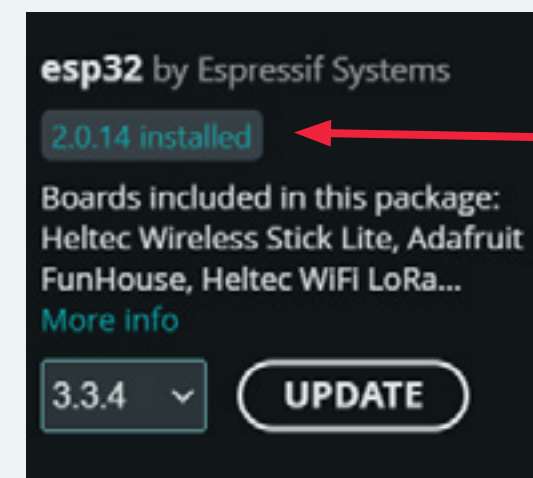
<https://github.com/matoslav/FerroWave>

Download Arduino IDE: <https://www.arduino.cc/en/main/software>

Install ESP32 board support in Arduino IDE

- Open Arduino IDE
- Go to File - Preferences
- Add to “Additional Board URLs”: https://espressif.github.io/arduino-esp32/package_esp32_index.json
- Go to Tools - Board - Boards Manager
- Search for “ESP32” and install “esp32” by Espressif Systems

IMPORTANT: Do not select the latest version! Use version 2.0.14 from the dropdown menu!



- **Select Board:** Tools - Board - ESP32 Arduino - ESP32 Dev Module

UPDATED IN v1.1

Use ESP32 core 2.0.14 AND set Partition Scheme to "Huge App" (Tools > Partition Scheme). If the sketch won't fit or won't boot, this is usually why.

Required Libraries

1. AudioTools (by Phil Schatzmann)

Primary audio processing library

GitHub: <https://github.com/pschatzmann/arduino-audio-tools>

2. ESP32-A2DP (by Phil Schatzmann)

Bluetooth A2DP audio sink

GitHub: <https://github.com/pschatzmann/ESP32-A2DP>

3. Adafruit NeoPixel

WS2812B LED control

GitHub: https://github.com/adafruit/Adafruit_NeoPixel

You can also download all the required libraries from the .zip file on GitHub:

<https://github.com/matoslav/FerroWave/blob/main/firmware/libraries.zip>

Just extract the contents to (on Windows the default is: C:\Users\Username\Documents\Arudino\libraries)

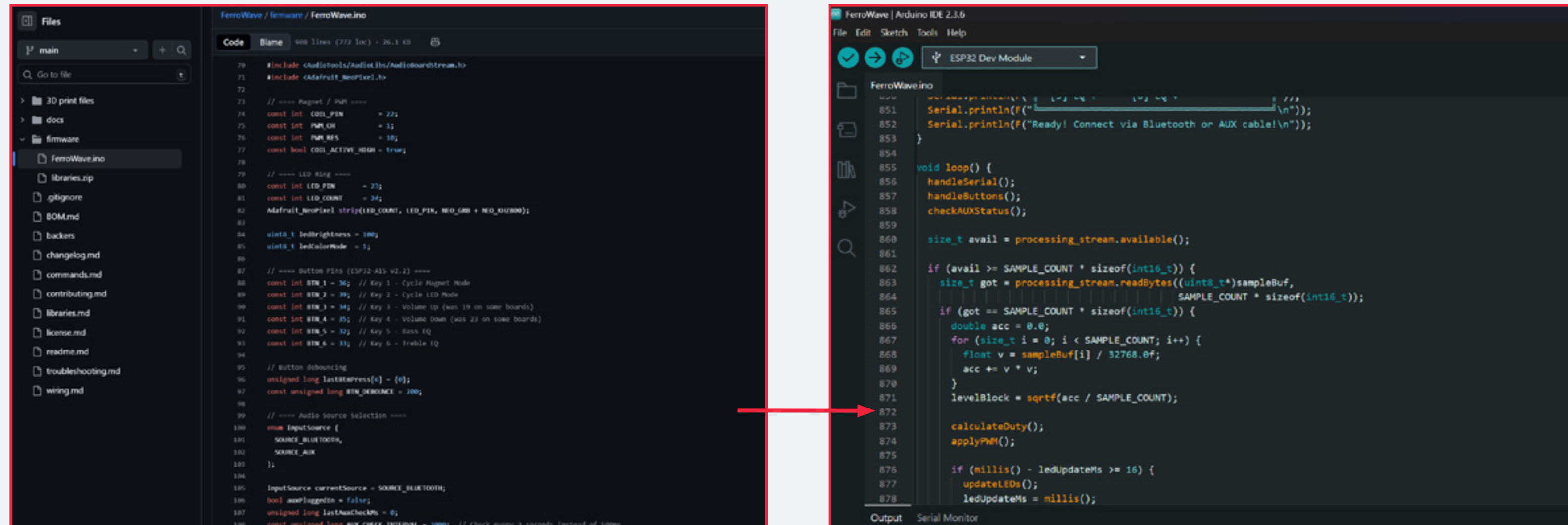
Also install a 4th library:
audio-driver (by pschatzmann)
github.com/pschatzmann/arduino-audio-driver
Some builds fail to compile without it.
All four are in firmware/libraries.zip.

3

PROGRAMMING

3.1

UPLOAD THE SKETCH



Next step is to copy-paste the FerroWave.ino from the GitHub into your sketch and upload it to the board: <https://github.com/matoslav/FerroWave/blob/main/firmware/FerroWave.ino>

UPDATED IN v1.1 - IMPORTANT

Recommended: flash FerroWave_fixed.ino instead.
It fixes repeating/stuck buttons, a strobe-mode bug,
and a pin conflict. Same GitHub firmware folder:
github.com/matoslav/FerroWave (firmware/)
See the Firmware Update pages at the end of this manual.

Take the USB cable used for programming out. Connect the power cables as they should be (only powering the ESP32 board via the other Micro-USB that's connected to our rail. Use Serial Monitor commands to try a few things out and play around with the parameters.

Magnet Modes (1-8)

Simply type the number:

- `1` = SMOOTH - Gentle flowing waves
- `2` = SPIKE - Sharp transients, dramatic spikes
- `3` = BOUNCE - Rhythmic pumping
- `4` = CHAOS - Aggressive, unpredictable
- `5` = PULSE - Beat-synced pulses
- `6` = WAVE - Slow building surges
- `7` = TREMOLO - Rapid flutter effect
- `8` = BREATH - Meditative breathing

LED Effects (c1-c10)

Type 'c' followed by the number:

- `c1` = Rainbow spin
- `c2` = Spectrum analyzer
- `c3` = Pulse (blue)
- `c4` = VU-Meter
- `c5` = Bass glow
- `c6` = Fire effect
- `c7` = Ocean waves
- `c8` = Strobe flash
- `c9` = Color chase
- `c10` = Sparkle

Fine-Tuning Parameters

Type letter + number (e.g., `f500`, `s120`):

- `f<num>` = PWM frequency (1-5000 Hz) - Try f4, f500, f1000
- `s<num>` = Sensitivity (0-200%) - How reactive to music
- `a<num>` = Attack speed (0-100) - How fast it responds
- `r<num>` = Release speed (0-100) - How fast it relaxes
- `d<num>` = Max duty (10-100%) - Power ceiling for safety
- `b<num>` = Base duty (0-50%) - Minimum always-on power
- `p<num>` = Spike intensity (0-100) - Extra boost for transients
- `l<num>` = LED brightness (0-255)
- `v<num>` = Volume (0-100%)

Audio Source Control

- `aux` = Switch to AUX input
- `bt` = Switch to Bluetooth

Utility Commands

- `?` = Show all current settings
- `t` = Test pulse (2 second max power test)
- `m` = List all magnet modes
- `n` = List all LED modes

UPDATED IN v1.1

If buttons repeat on their own:

- 1) Flash FerroWave_fixed.ino
 - 2) Set DIP switch position 2 = ON (rest OFF)
 - 3) Move LED ring data wire to GPIO 21
- Full details: Firmware Update section (end).

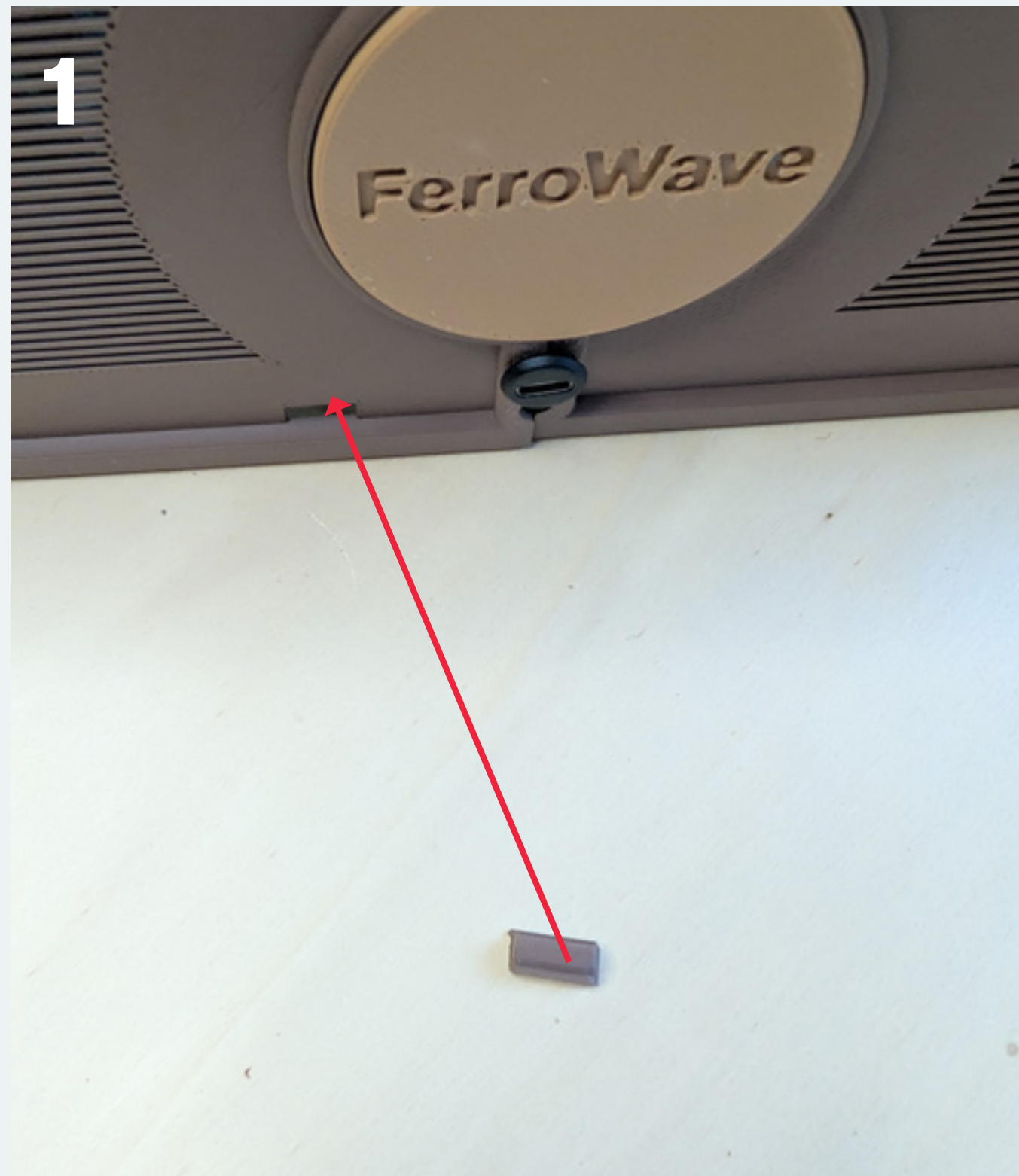


4

FINISHING TOUCHES

3.1

BACK PANEL



The Battery Pack need waking before use (if anybody finds a better solution...) so this is why we have this Wake Up button at the back.
Slide it from the back then secure the Back Plate with four M3x3mm screws.
That's IT!



Now it's up to you! Build it, experiment with it, share you progress, either via GitHub or MakerWorld's Crowdfunding campaign.

GitHub: github.com/matoslav/FerroWave | MakerWorld: makerworld.com/en/crowdfunding/70-ferrowave

What changed and why

Some builders saw buttons repeat or stick on their own, most often the magnet-mode and LED-mode buttons. It happens even with the buttons untouched and on a fresh board, so it is a firmware issue, not your wiring or soldering.

Some buttons were assigned to ESP32 pins that cannot hold a steady "not pressed" signal on this board, so they drifted and registered phantom presses. The fix moves the buttons to the correct on-board KEY pins and makes the button reader noise-proof.

The fix, in three steps

- 1) **Flash FerroWave_fixed.ino** - Download and flash this file from the GitHub firmware folder, instead of the original FerroWave.ino.
- 2) **Set DIP switch position 2 = ON** - (all the other DIP switches OFF). This connects the on-board KEY buttons.
- 3) **Move the LED ring data wire to GPIO 21** - The fix uses the old LED pin for a button, so the ring's data wire moves.

GitHub is the home base: github.com/matoslav/FerroWave

Firmware, troubleshooting and discussion all live there. Found something? Open an issue so everyone benefits.

- **Buttons repeat or get stuck on their own**

Flash FerroWave_fixed.ino, set DIP switch position 2 = ON (rest OFF), and move the LED ring data wire to GPIO 21. After flashing, open the Serial Monitor and watch it without touching anything - it should stay quiet. If it still scrolls button presses, re-check the DIP switch.

- **Sketch won't compile (missing driver)**

Install the audio-driver library by pschatzmann (github.com/pschatzmann/arduino-audio-driver). All four libraries are also bundled in firmware/libraries.zip on GitHub.

- **Sketch won't fit or board won't boot after upload**

Make sure ESP32 core is version 2.0.14 and Partition Scheme is set to "Huge App" (Tools > Partition Scheme).

- **Strobe LED effect (c8) does nothing**

Fixed in FerroWave_fixed.ino - it now flashes on the beat instead of staying dark.

- **Button 1 still acts up (rare)**

Button 1 sits on a special input-only pin. The new firmware guards it in software, but if needed you can add a 10k resistor from GPIO 36 to 3.3V as a belt-and-suspenders fix.

Huge thanks to the community builders on GitHub who found and confirmed these fixes, and to everyone on MakerWorld for your patience and build reports.